GUEST LECTURE | DATA SCIENCE FOR HEALTH AND BIOMEDICAL SCIENCES

Promise and Pitfalls: Data Science and AI in Medicine

From routine data to clinical decisions: evidence, bias and ethics

Dr Syed Ahmar Shah

The University of Edinburgh | Usher Institute

School of Population Health Sciences
USHER INSTITUTE

WHERE WE ARE

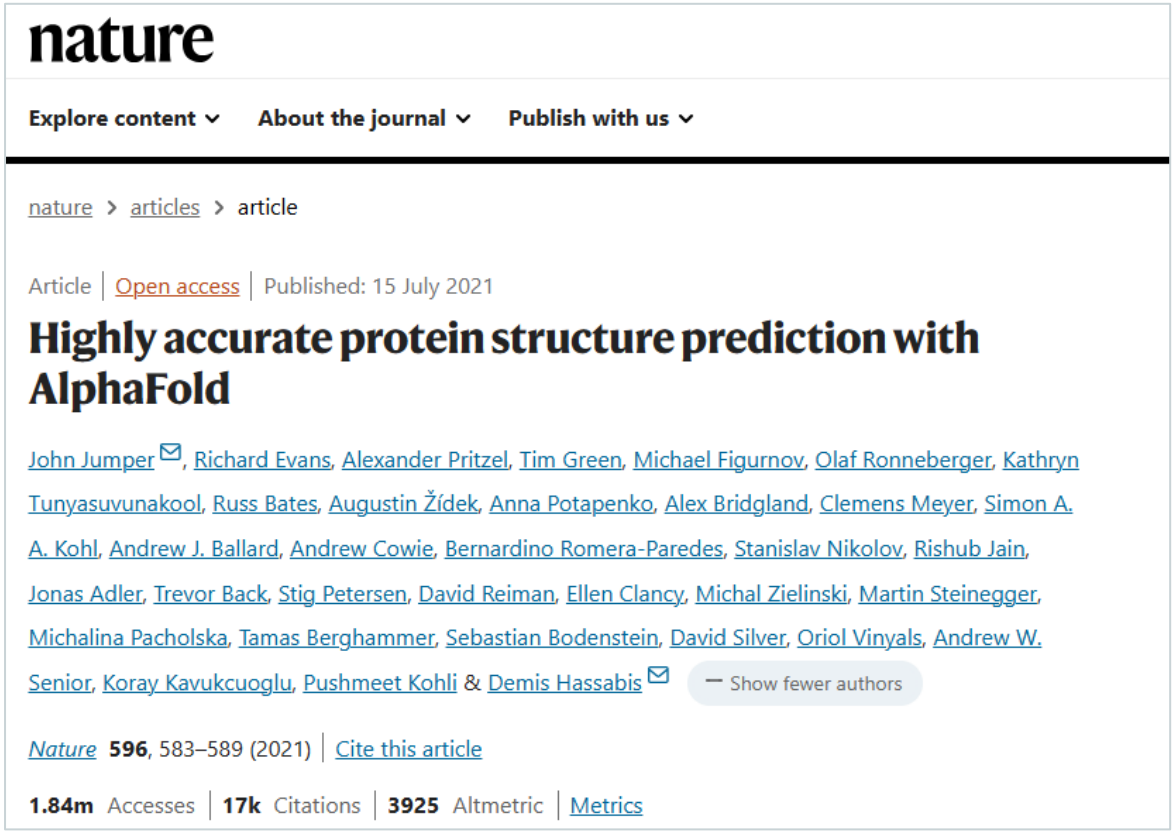
AI is already changing science

Two landmark examples, both recognised with 2024 Nobel Prizes



Geoffrey Hinton

Foundations of deep learning



nature

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Article | Open access | Published: 15 July 2021

Highly accurate protein structure prediction with AlphaFold

John Jumper ✉, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, Alex Bridgland, Clemens Meyer, Simon A. A. Kohl, Andrew J. Ballard, Andrew Cowie, Bernardino Romera-Paredes, Stanislav Nikolov, Rishub Jain, Jonas Adler, Trevor Back, Stig Petersen, David Reiman, Ellen Clancy, Michal Zielinski, Martin Steinegger, Michalina Pacholska, Tamas Berghammer, Sebastian Bodenstein, David Silver, Oriol Vinyals, Andrew W. Senior, Koray Kavukcuoglu, Pushmeet Kohli & Demis Hassabis ✉ — Show fewer authors

Nature **596**, 583–589 (2021) | Cite this article

1.84m Accesses | **17k** Citations | **3925** Altmetric | Metrics

AlphaFold: highly accurate protein structure prediction

Jumper et al., Nature, 2021

ABOUT ME

Almost two decades at the interface of medicine and engineering



1

Engineering first

First degree in Electronics Engineering

2

11 years in Oxford

MSc and PhD in Biomedical Engineering, then postdoctoral research

3

8 years in Edinburgh

Group leader: health data, AI and digital health

Data-driven Innovation in Medicine · Edinburgh

AI4Health
Bridging Health and AI: Empowering Innovators

ROADMAP

What we will cover today

1 **Setting the stage**
From Medicine 1.0 to 4.0, and the digital transformation of health and social care

2 **A primer on AI, data science and machine learning**
Key terms, types of learning and how we measure a model

3 **Case studies from my own research**
Patient, population and health-system scales

4 **Can we trust it? Data, evidence and ethics**
Bias, validation, the AI chasm and large language models

5 **The future and take-home messages**
Opportunities, challenges and what to ask of any AI claim

ROADMAP

By the end of this lecture you should be able to

Place AI in context

Explain how medicine has moved from observation towards data-driven prevention, and why that matters now.

Use the vocabulary

Distinguish AI, machine learning and data science, and describe supervised, unsupervised and reinforcement learning.

Read the numbers

Interpret sensitivity, specificity, PPV, AUROC and calibration, and explain why prevalence changes what a positive result means.

Question the claims

Recognise how bias enters through data, labels and deployment, and ask better questions of any AI claim.

From the real world to impact

We will follow this chain: a failure at any link limits the benefit to patients

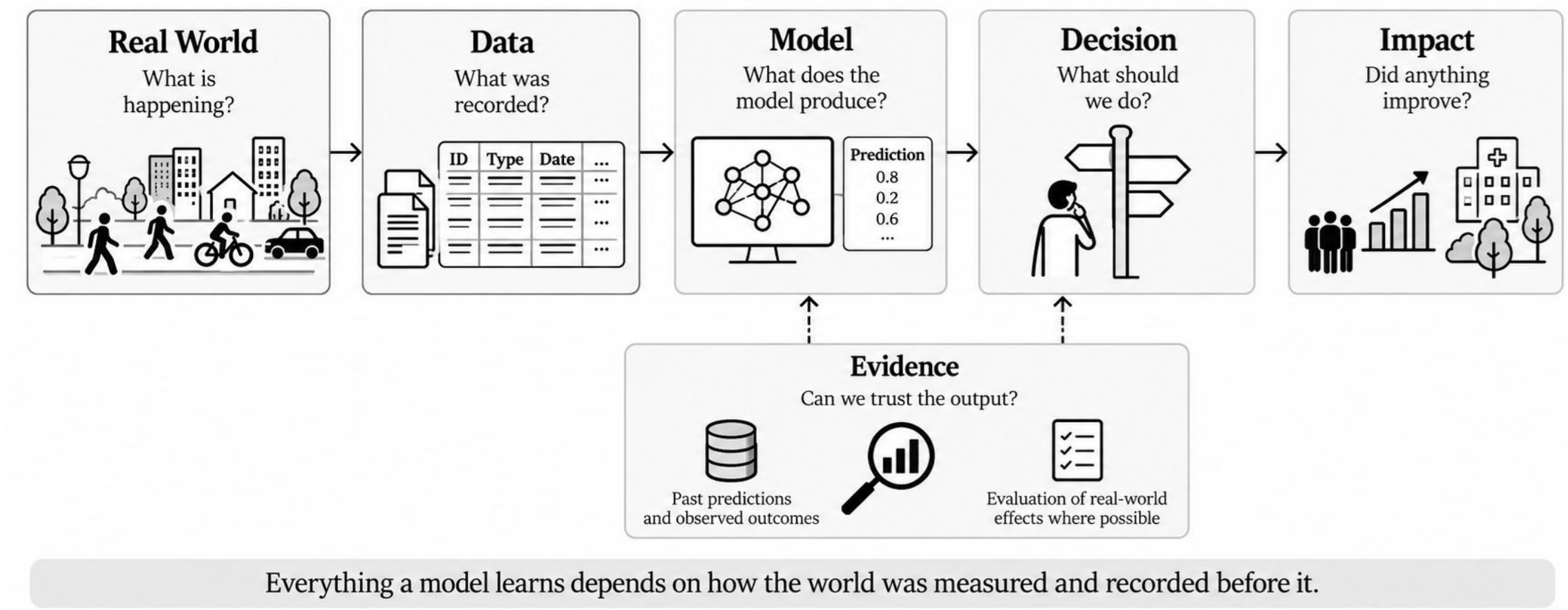


Figure from: Shah SA. Understanding AI for the Real World (in preparation)

1

Setting the stage

From Medicine 1.0 to Medicine 4.0, and the digital transformation of health and social care

Four eras of medicine

Medicine 4.0 moves from treating disease to predicting and preventing it



Medicine 1.0

5th century BC to 18th century

Observation and reasoning

Hippocrates: observe the patient, not superstition



Medicine 2.0

18th to mid-20th century

Revolution and discovery

Anaesthesia, germ theory, vaccines and antibiotics



Medicine 3.0

Mid-20th century to today

Technology and specialisation

Imaging, genetics, transplants, life-saving drugs



Medicine 4.0

Emerging now

Prevention and personalisation

Predict risk before disease develops

Data science, AI and machine learning are what make Medicine 4.0 possible: prediction needs data, models and evidence that they work.

From observation to intervention: Medicine 1.0 and 2.0



Medicine 1.0: the dawn of scientific medicine

5th century BC to 18th century

- Mindset: from superstition to observation and reasoning
- Hippocrates put observation before superstition
- Treatments based on balancing bodily fluids
- Surgery was risky and painful, with high mortality



Medicine 2.0: revolution and discovery

18th century to mid-20th century

- Anaesthesia: painless and more complex surgery
- Germ theory: hygiene and sanitation
- Vaccines and antibiotics: smallpox prevented, bacterial infection treated
- Infectious diseases eradicated or contained

Medicine 3.0: technology and specialisation

Mid-20th century to the present



Technological progress

Imaging such as X-rays and CT scans, genetics and diagnostics

New drugs and treatments

Life-saving medicines and complex procedures such as organ transplants

Specialisation

Deeper expertise in specific areas of medicine

COVID-19 vaccines

A recent example of rapid technological advance

Medicine 4.0: prevention, personalisation, healthspan

A shift in mindset as much as in technology

Prevention

Greater emphasis on preventing disease before it develops

Personalised medicine

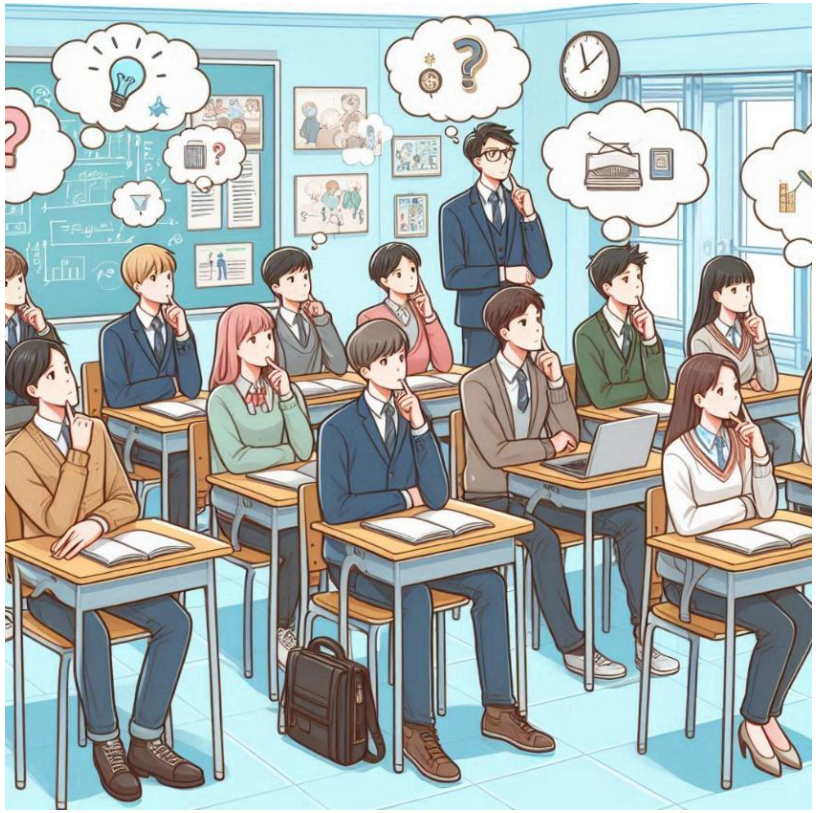
Treating each patient as a unique individual

Healthspan over lifespan

High quality of life until death, not simply a longer life

Resilient systems

Healthcare systems that manage health proactively, not only treat illness



Why do we need data science now?

Ageing populations, chronic disease and growing demand for healthcare



The United Nations Agenda for Sustainable Development

People are living longer

More people live with chronic disease

Demand for care keeps growing

Data-driven technologies can help make healthcare systems sustainable, if we use them well.

Health and social care: a bird's-eye view

Every setting generates data, and each answers a different kind of question



Primary care

First point of contact, such as the GP

Secondary care

Hospital and specialist services

Tertiary care

Highly specialised centres

Public health

Prevention and the health of populations

Community services

Care closer to home

Social care

Support with daily living and care at home

The case studies in Part 3 are chosen from across these settings.

What happens between healthcare encounters?

Digital health changes the nature of the data we can collect

Routine healthcare data

- Consultations and diagnoses
- Prescriptions and tests
- Admissions

Digitally generated data

- Symptoms and physiology
- Activity and repeated measurements
- Continuous, high-frequency data

A richer longitudinal view of the patient

The challenge changes from having too little information to identifying what is clinically meaningful.

Health data can inform decisions at three scales

Different questions need different data, methods and evidence



PATIENT

Who is at risk?

Prediction and decision support



POPULATION

What can we learn?

Disease, treatment and outcomes



HEALTH SYSTEM

What does the system need?

Capacity, disruption and recovery

Different questions require different data, methods and evidence.

2

A primer on AI, data science and machine learning

The few ideas that connect the terminology, and how we judge whether a model works

What is artificial intelligence?

Machines doing tasks that usually need human intelligence



Learning from experience

Predictions improve as more patient records arrive



Understanding language

Summarising a clinic letter



Recognising patterns

Spotting a fracture on an X-ray



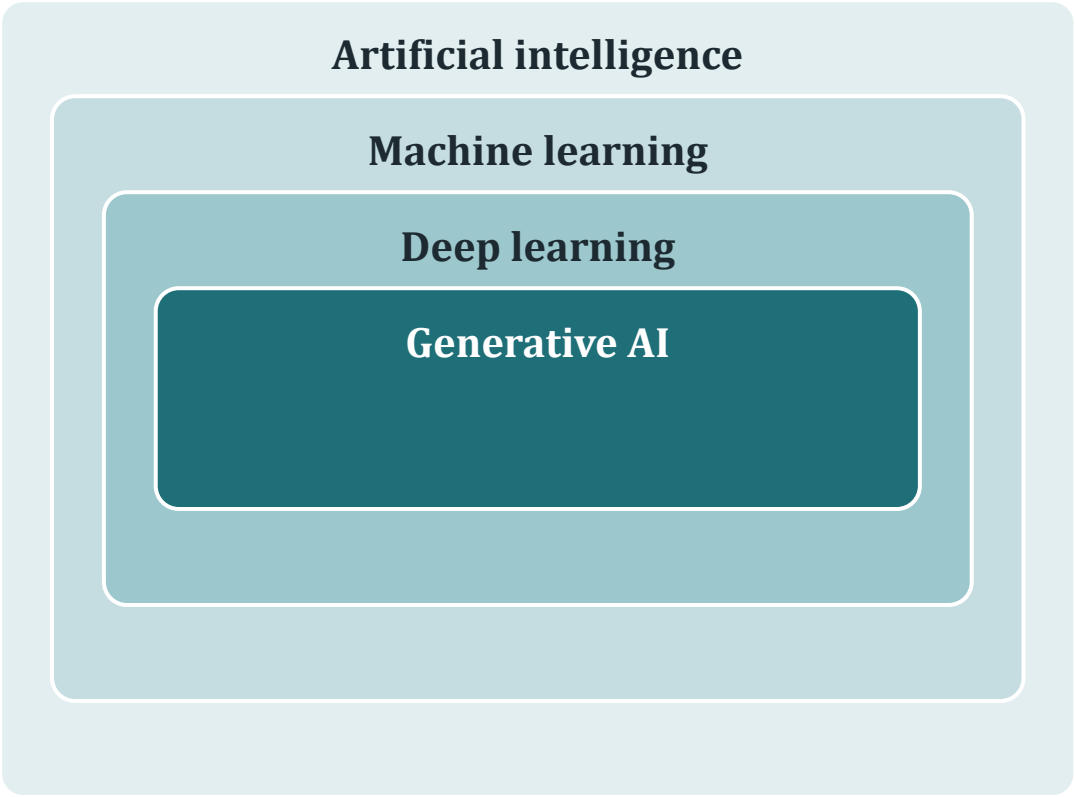
Solving problems

Planning operating theatre schedules

AI does not follow step-by-step instructions for each task: it learns how from examples.

The AI landscape

Machine learning is the dominant approach in AI today



Artificial intelligence

Any technique that lets machines do tasks needing human intelligence, including hand-written rules.

Machine learning

Systems that learn patterns from data rather than following written rules.

Deep learning

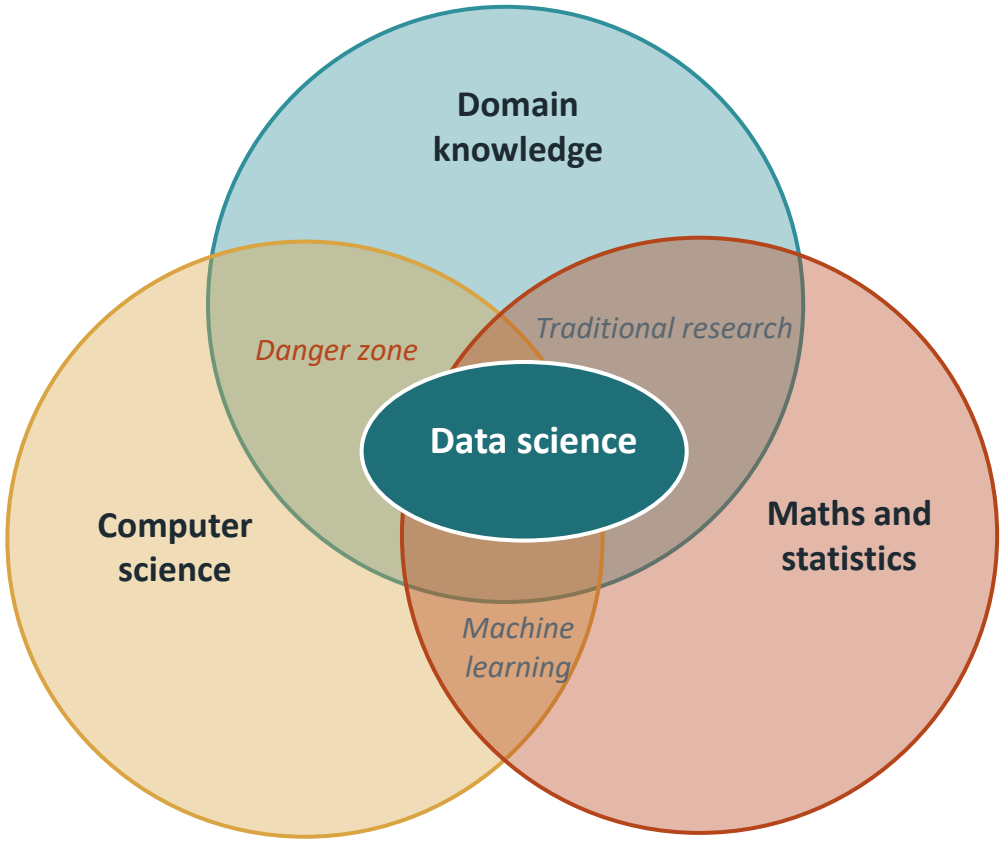
Machine learning with many-layered neural networks; behind modern imaging and speech tools.

Generative AI

Deep learning models that create new text, images or code, such as large language models.

Data science vs machine learning

Machine learning builds methods; data science applies them to real problems. The boundary is blurry



Machine learning

Methods that learn from data, from a fitted line (slope and intercept learnt from data) to large neural networks. Some people develop the methods; others apply them; some do both.

Data science

The broader term: using such methods on real problems in a domain. Rarely plug and play, so people adapt or create methods to fit real-world data constraints.

Danger zone

Coding skill and domain knowledge without statistics: results that look right and are not.

These are not hard boundaries: the same person can develop methods and apply them.

Adapted from Drew Conway's Data Science Venn Diagram (2010)

What is machine learning?

Machine learning learns the rules from examples instead of being given them

Task: decide whether an email is spam

Approach 1

Write the rules

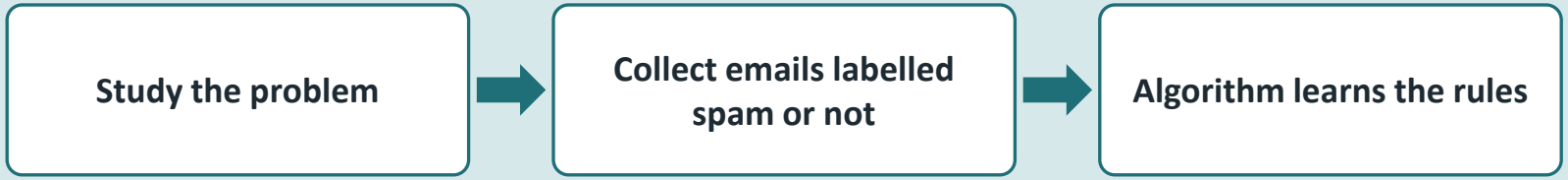
Breaks when spammers change tactics; people must keep rewriting the rules.



Approach 2

Learn the rules

Adapts by retraining on new examples.



Machine learning: learning from data without being explicitly programmed (Arthur Samuel, 1959)

Three types of machine learning

What you give the model decides what it can learn

Supervised

YOU PROVIDE

Inputs with known outcomes

IT LEARNS

To predict the outcome for new cases

IN HEALTHCARE

Flagging COPD flare-ups from vital signs

Unsupervised

YOU PROVIDE

Inputs without outcomes

IT LEARNS

Groups and structure in the data

IN HEALTHCARE

Finding patient subgroups with similar symptoms

Reinforcement

YOU PROVIDE

Actions, and rewards for good outcomes

IT LEARNS

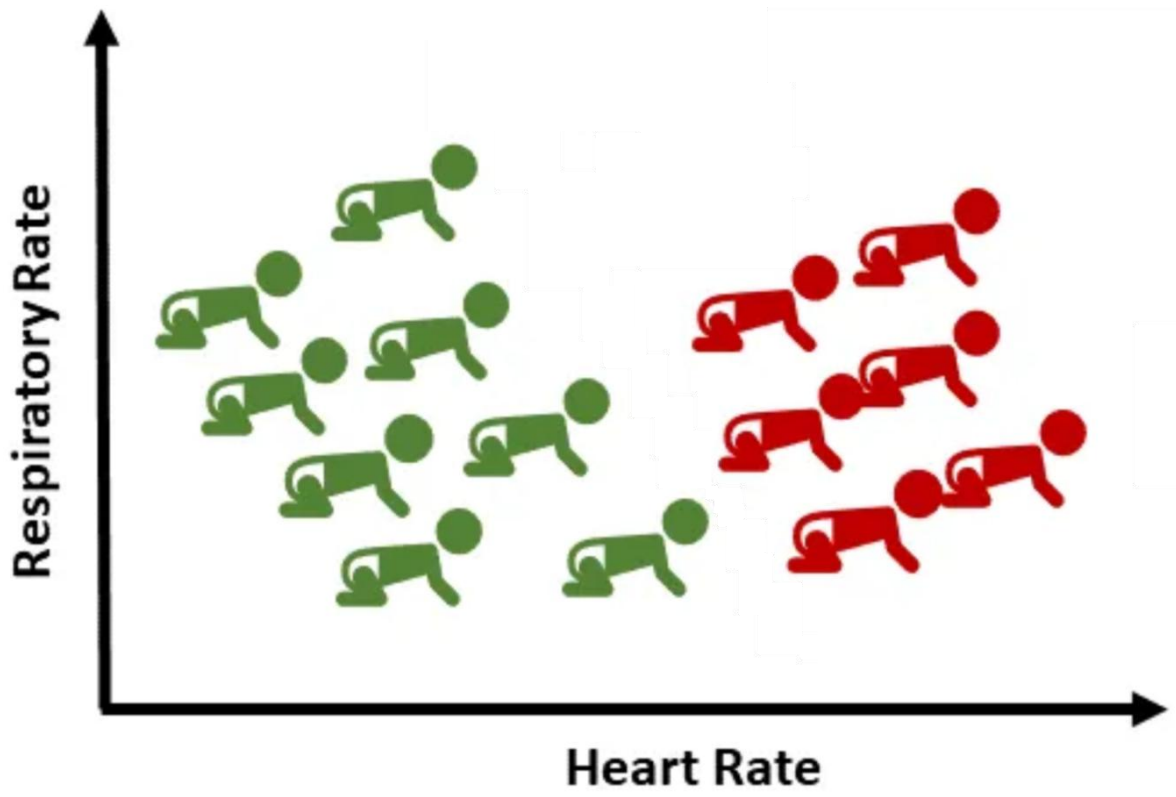
Which actions lead to the best result

IN HEALTHCARE

Choosing fluid and drug doses in sepsis (Komorowski 2018)

Supervised learning

The model learns a boundary from labelled examples, then applies it to new cases



1

Give labelled examples

Heart and breathing rates of babies already known to be healthy or sick.

2

Learn the boundary

The model finds the line that best separates the two groups.

3

Predict new cases

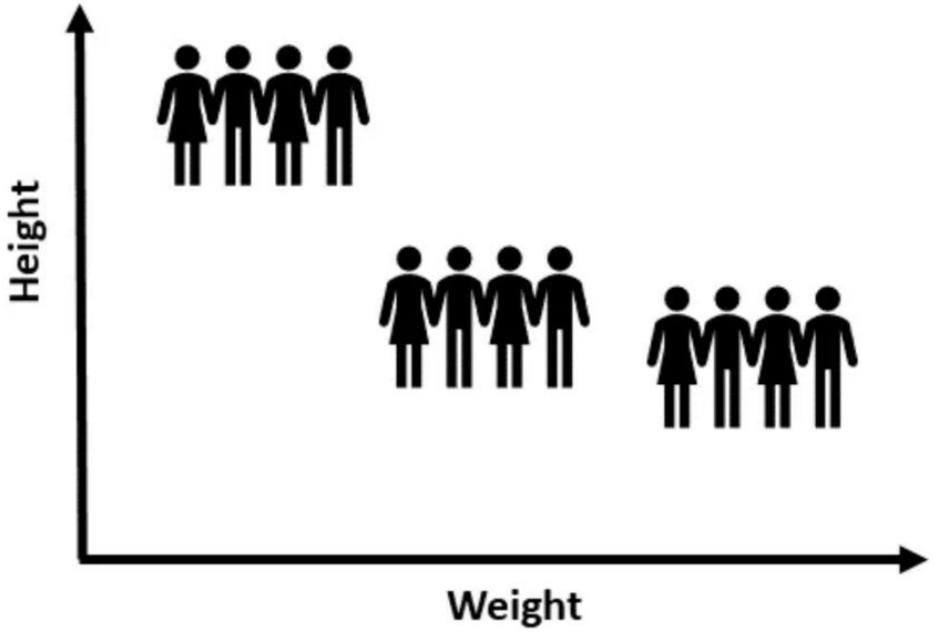
A new baby's readings fall on one side of the line.

In healthcare: Case Study 1 labels each week as stable or before a flare-up, then learns to tell them apart.

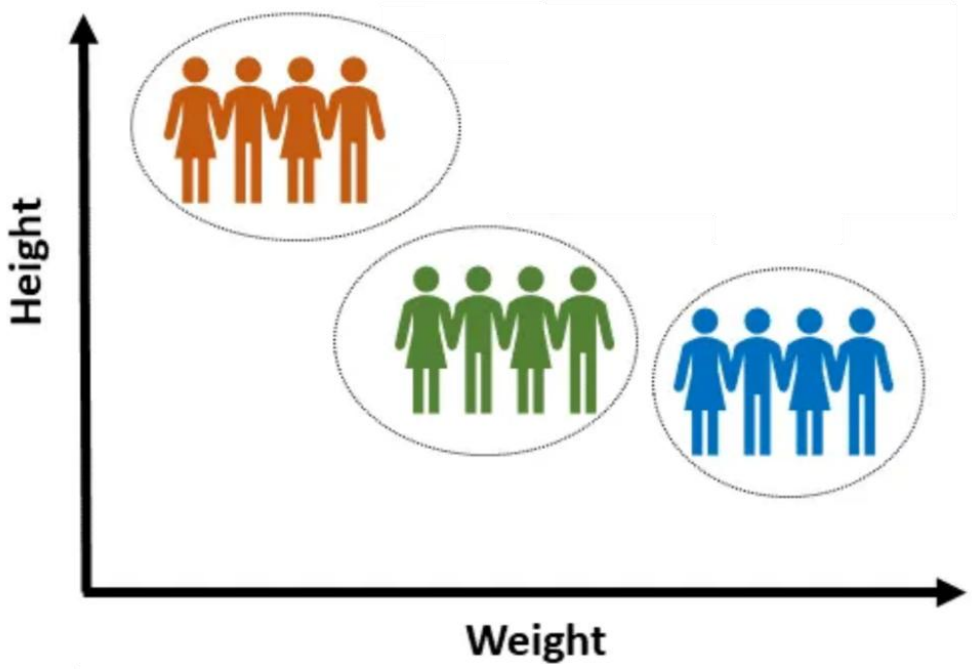
Unsupervised learning

With no labels, the model finds groups on its own; people decide what they mean

Input: measurements, no labels



Output: three groups found by the model



The model only finds similarity. Naming the groups and testing whether they matter is a human job.

Is learning from data a new idea?

A caveman, Isaac Newton and a data scientist round a fire



OBSERVE

Caveman

Watched wildfires, found that friction makes fire, built a bow drill and passed the method on.

MODEL

Isaac Newton

Collected observations, proposed a hypothesis to explain them, then used it to build and predict.

COMMUNICATE

Data scientist

Learns from data using models and communicates the results.

The loop is the same in all three. What is new is the amount of data and the computing power.

Adapted from: Shah A. Learning from Data: A Bird's Eye View. Towards Data Science (Medium), Jan 2021.

What is driving machine learning now?

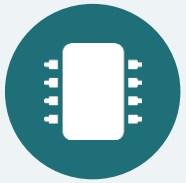
The ideas are old; data and computing power at today's scale are new



NEW IN HISTORY

Data

Electronic health records, medical imaging, genomics, wearable sensors and digital platforms generate more data than ever.



NEW IN HISTORY

Computing power

GPUs, specialised AI accelerators and cloud computing train large models in days.



MADE POSSIBLE BY BOTH

Algorithms

Deep networks, transformers and self-supervised learning turn data and computing into models.

The ideas behind learning from data are old. The data and the computers to learn from it at scale are what changed.

From task-specific ML to foundation models

Traditional, task-specific ML

Clinical measurements and labelled outcomes

Train a model for a defined task

Predict exacerbation risk

VS

Foundation models

Large-scale pretraining on extensive datasets

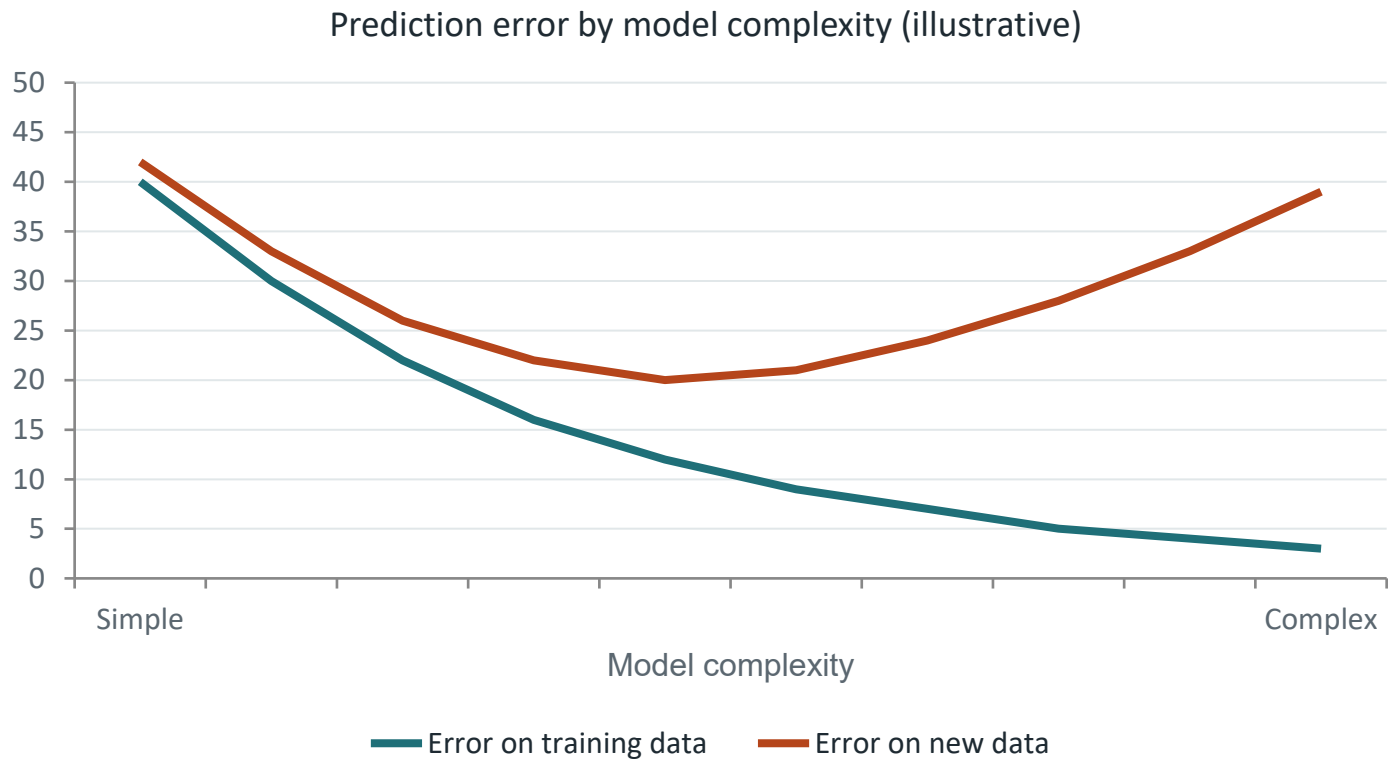
Reusable model adapted to different tasks

Summarise, answer, classify or generate

Conceptual comparison: foundation models may also classify and predict, and some add supervised or reinforcement-learning stages.

Generalisation and overfitting

Judge a model on data it has not seen, not the data it learned from



Memorising is not learning

A model that fits its training data perfectly can fail on new patients.

Hold data back

Split into training, validation and test sets before modelling starts.

Test somewhere new

A different hospital or a later year is the hardest test.

The confusion matrix

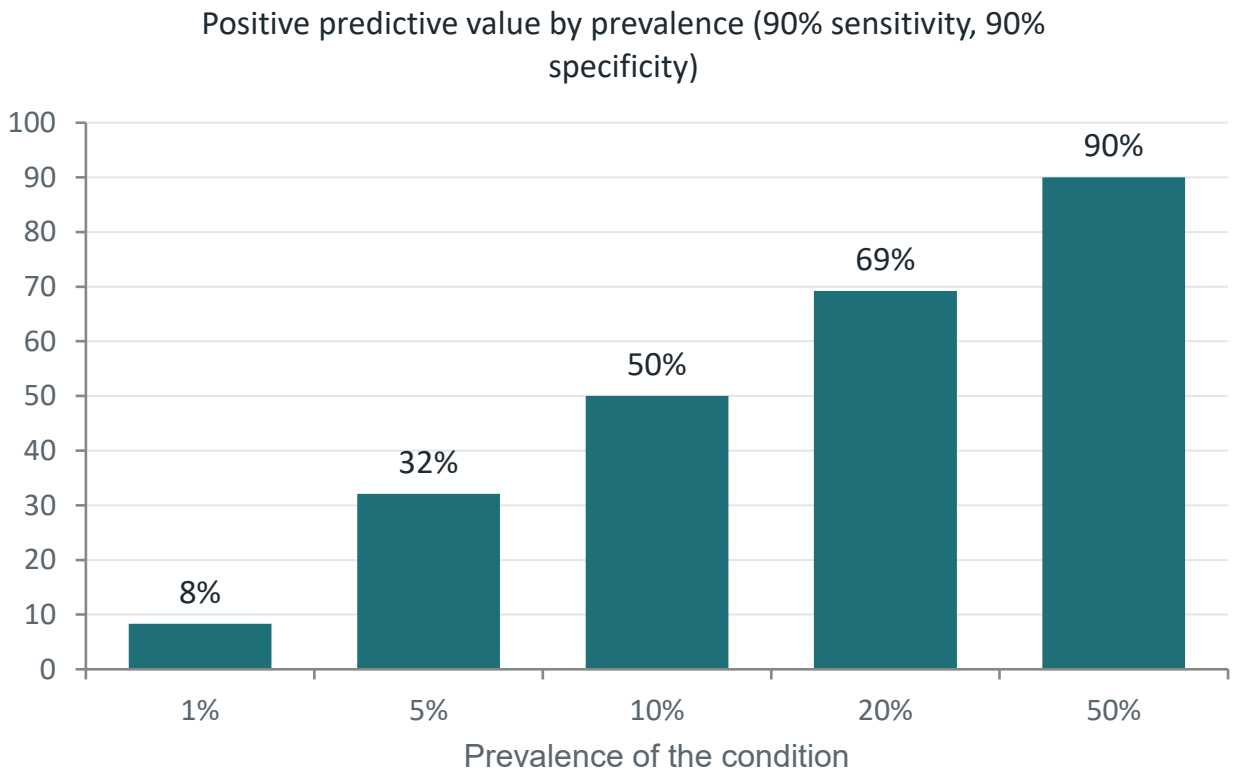
Every classification metric is built from the same four counts

	EVENT HAPPENED	EVENT DID NOT HAPPEN
Model flags the patient	**True positive (TP)** Flagged, and the exacerbation followed	**False positive (FP)** Flagged, but nothing happened: a false alarm
Model does not flag	**False negative (FN)** Not flagged, but the exacerbation happened: a miss	**True negative (TN)** Not flagged, and the patient stayed stable

The two errors rarely cost the same. In COPD home monitoring, a false alarm costs a nurse phone call; a miss can mean a hospital admission.

Why prevalence matters

At 1% prevalence, a 90%-accurate test raises 11 false alarms for every true case



Worked example

- Screen 1,000 people; 10 have the condition
- 9 are correctly flagged
- 99 of the 990 healthy people are also flagged
- **PPV = 9 ÷ 108 = 8%**
- Fewer than 1 in 12 flagged patients has the condition

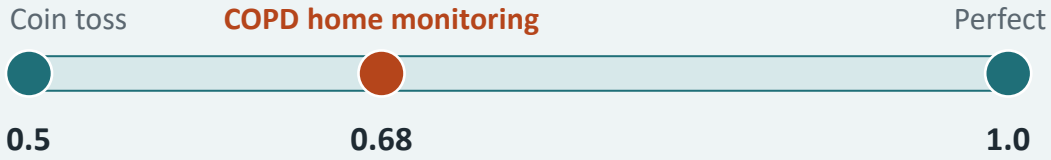
Test assumed to have 90% sensitivity and 90% specificity. PPV = sens × prev ÷ [sens × prev + (1 − spec) × (1 − prev)]

Discrimination vs calibration

AUROC shows whether a model ranks patients well, not whether its risks are right

Discrimination (AUROC)

Does a patient who has the event score higher than one who does not?



Calibration

When the model says 20% risk, do about 20 in 100 such patients have the event?

20 of 100
have the event: calibrated

8 of 100
have the event: the model overstates risk 2.5-fold

A model can rank patients well and still overstate everyone's risk. Report both.

Models are imperfect, but collectively they tell a story

No single model or metric gives the full picture



A 120-year journey with 5,000+ passengers in hibernation

- Due to a system malfunction, one passenger wakes up 90 years too early
- As the journey progresses, multiple failures begin to happen
- No single failure tells the whole story; together they do

Every model and metric has limits; together they tell us whether to trust a model.

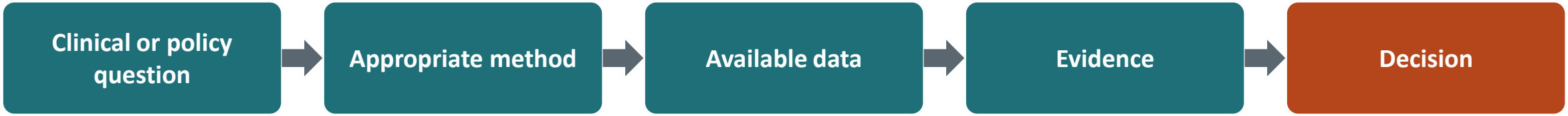
Clip: Passengers (2016)

3

Case studies from my own research

Published studies at the patient, population and health-system scales

The research question should determine the data and method

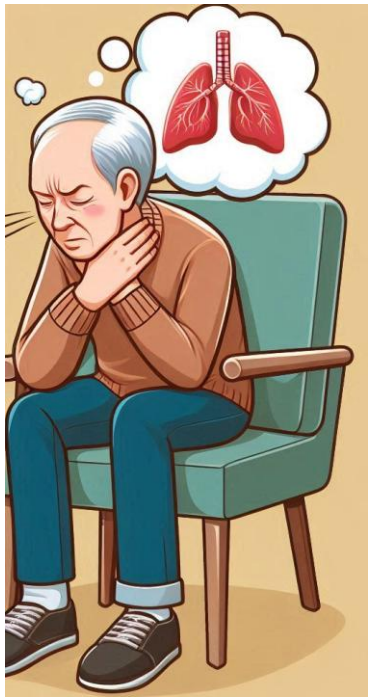


Methods you will meet in the case studies

Risk prediction and prognostic modelling	Survival and longitudinal analysis	Observational epidemiology
Interrupted time series	Forecasting and scenario modelling	Digital signal processing

More sophisticated modelling does not automatically mean more useful evidence.

Early warning of COPD exacerbations during home monitoring



- COPD is predicted to become the third leading cause of death by 2030
- Progressive chronic disease affecting the lungs
- Symptoms: cough, breathlessness, wheezing, chest tightness, fatigue and dizziness
- An acute worsening (an “exacerbation”) leads to hospitalisation if not treated promptly

JOURNAL OF MEDICAL INTERNET RESEARCH Shah et al

Original Paper

Exacerbations in Chronic Obstructive Pulmonary Disease: Identification and Prediction Using a Digital Health System

Syed Ahmar Shah[1,2], DPhil (Oxon); Carmelo Velardo[2], PhD; Andrew Farmer[3], DM; Lionel Tarassenko[2], DPhil (Oxon)

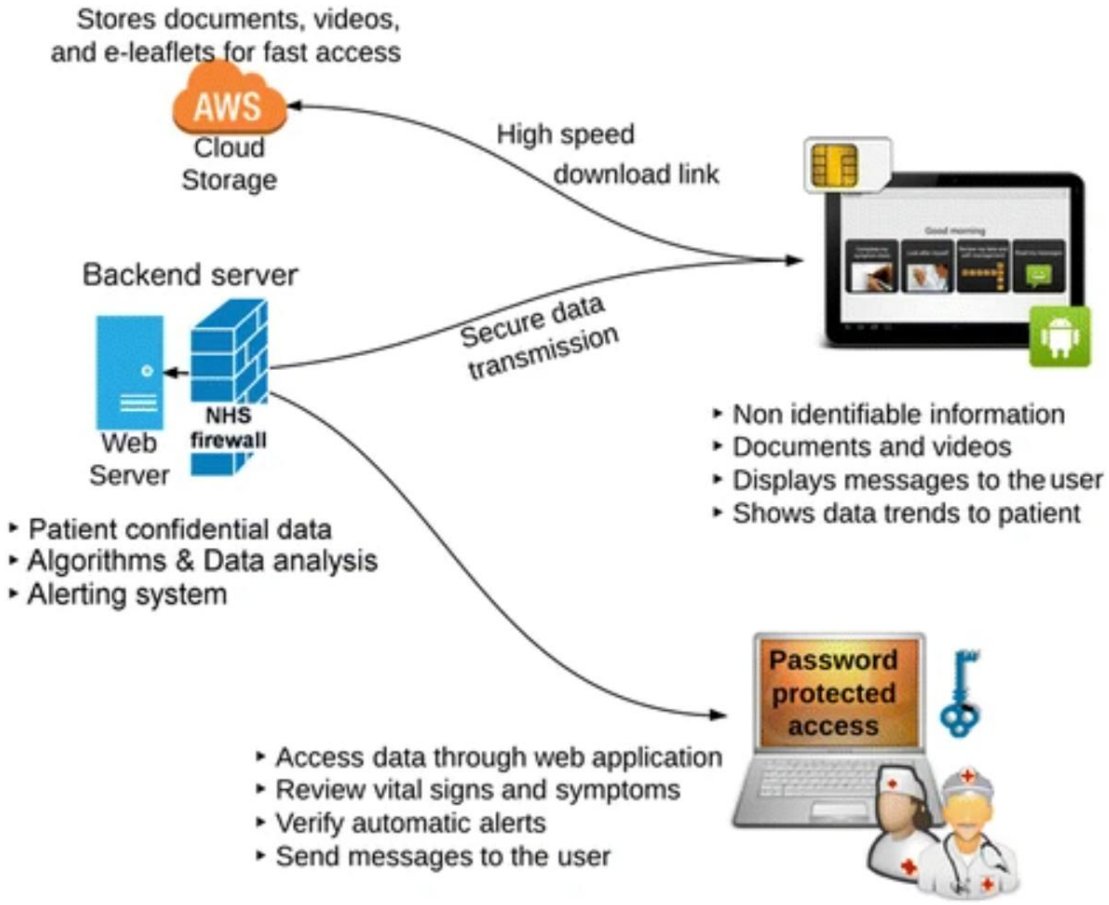
[1]Nuffield Department of Clinical Neurosciences, University of Oxford, Oxford, United Kingdom
[2]Institute of Biomedical Engineering, Department of Engineering Science, University of Oxford, Oxford, United Kingdom
[3]Nuffield Department of Primary Care Health Sciences, University of Oxford, Oxford, United Kingdom

DOI: 10.2196/jmir.7207

Objective: reduce episodes of exacerbation with digital health, improving quality of life.

Study set-up: COPD home monitoring

Patients used a mobile phone app and a pulse oximeter at home





Pulse oximeter: 30-second photoplethysmogram, heart rate and SpO_2

100+

patients in a randomised controlled trial

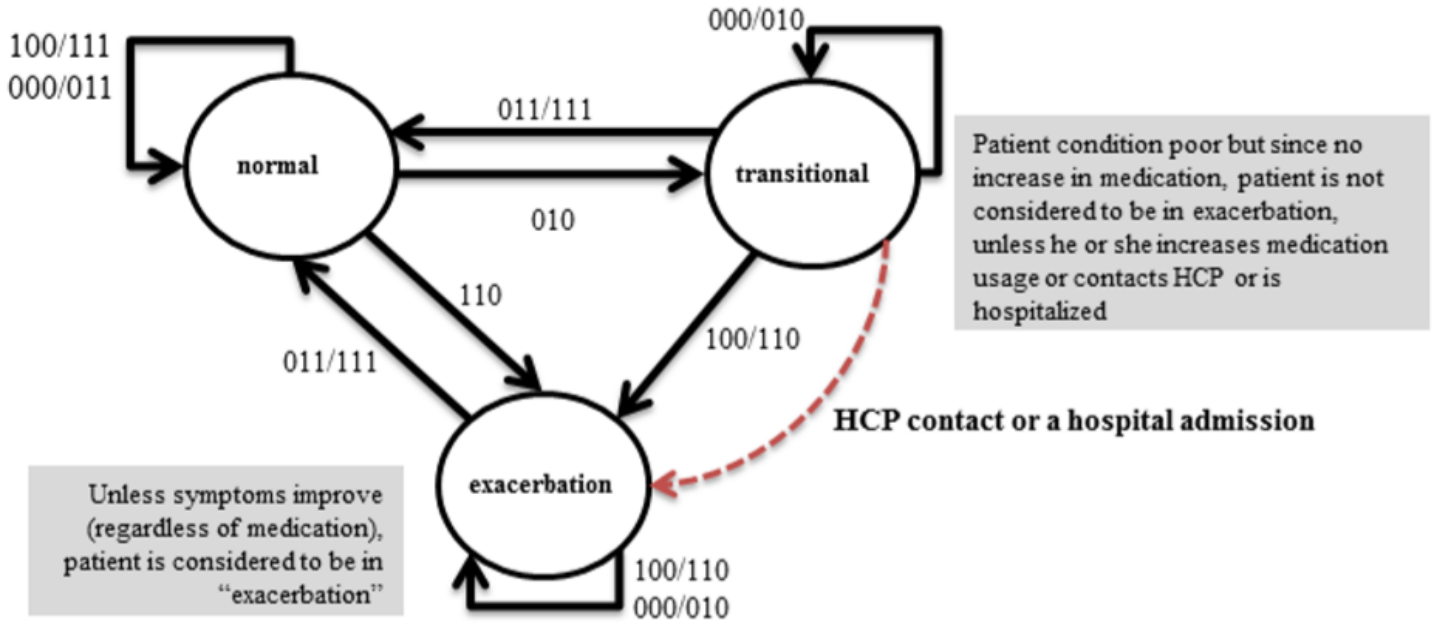
1 year

of longitudinal monitoring

Image first published in: Velardo C, Shah SA, et al. Digital health system for personalised COPD long-term management. BMC Med Inform Decis Mak, 2017;17:1-13

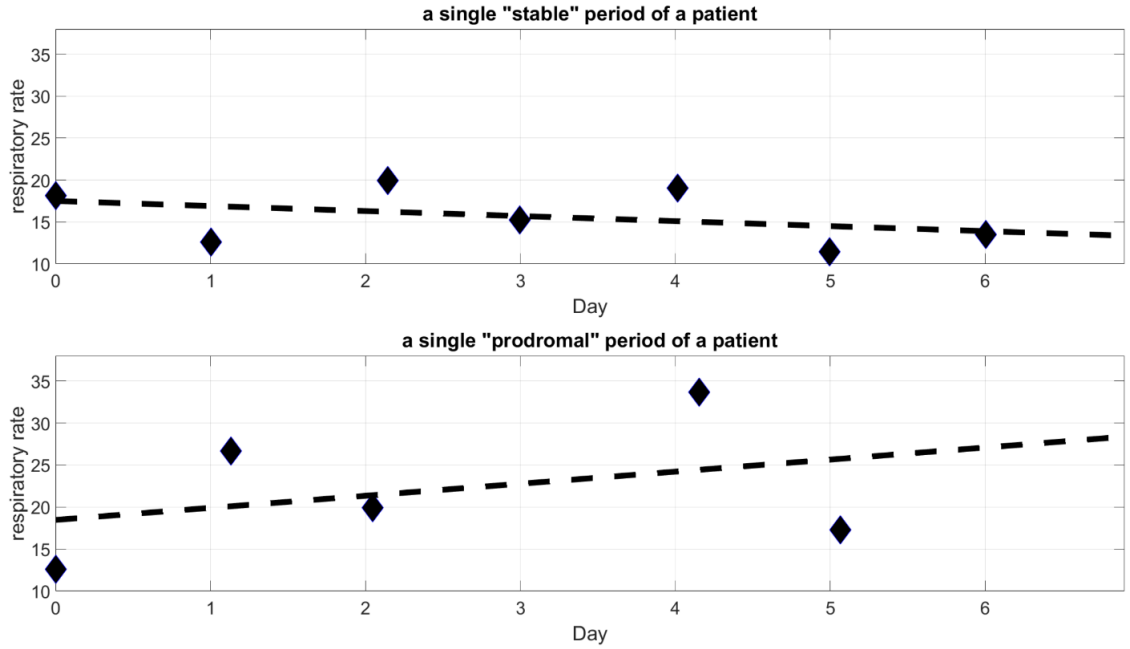
Rethinking the definition of an exacerbation

Before any model, we had to decide what counts as an exacerbation



Input	Interpretation	Input	Interpretation
000	no or usual medication, no change in symptoms	100	medication increase, no change in symptoms
010	no or usual medication, symptoms get worse	110	medication increase, symptoms get worse
011	no or usual medication, symptoms improve	111	medication increase, symptoms improve

Prodromal vs stable periods: a binary classifier



7-day sliding window

27,260

sessions recorded in the clinical trial (5.3 per week on average, for 12 months)

Inputs

- Respiratory rate
- Oxygen saturation
- Heart rate

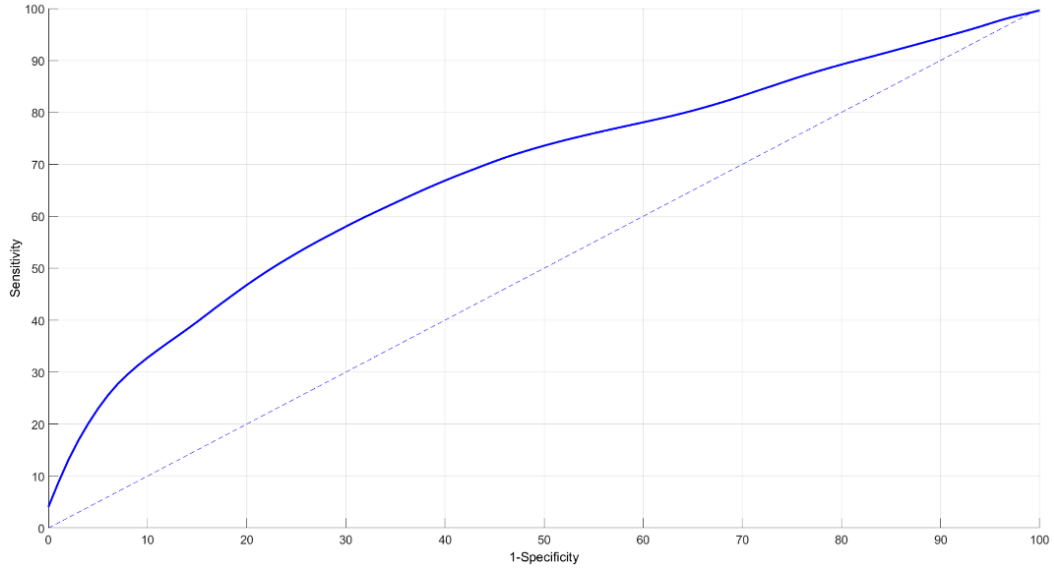
→ **Machine learning model (binary classifier)** →

Stable or Prodromal

Results: a real but modest early signal

Vital sign	Mean AUC[a] (95% CI)
Pulse rate (PR)	0.578 (0.578-0.578)
SpO_2[c]	0.658 (0.657-0.658)
Respiratory rate (RR)	0.605 (0.604-0.605)
PR + SpO_2	0.664 (0.664-0.664)
RR + PR	0.612 (0.612-0.612)
RR + SpO_2	0.672 (0.671-0.672)
RR + PR + SpO_2	0.682 (0.681-0.682)

Mean AUC by vital sign combination



Receiver operating characteristic curve

AUROC 0.68 with all three vital signs combined (SpO_2 alone 0.66, pulse rate alone 0.58).

Case study 1: summary

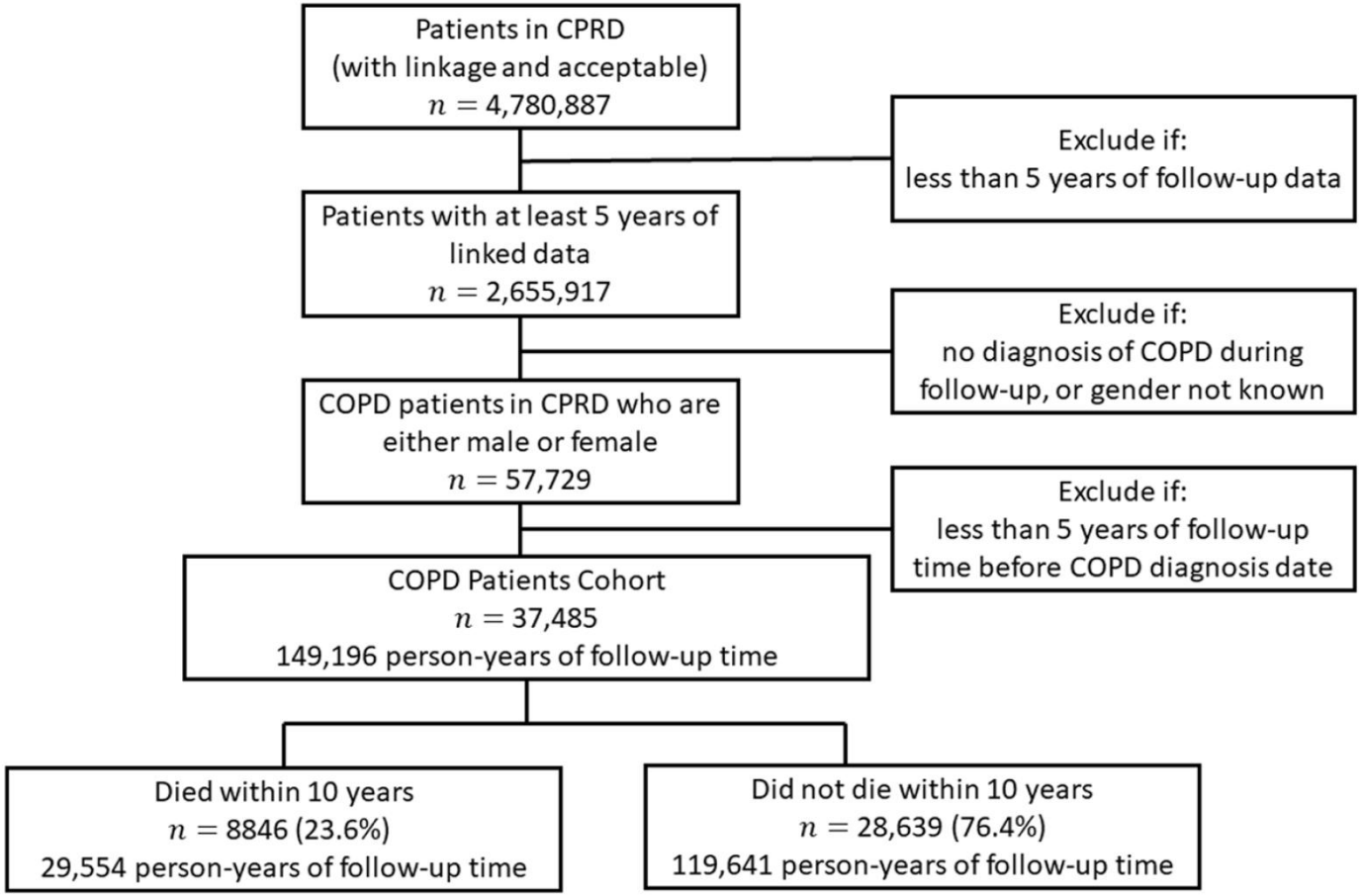
A real but modest early warning signal from home vital signs

PROBLEM	COPD exacerbations lead to hospital admission if not treated promptly. Can home vital signs warn early?
DATA	EDGE randomised controlled trial: 100+ patients monitored at home for 12 months, 27,260 sessions of SpO_2, pulse rate and respiratory rate
MODEL	Exacerbations redefined from symptoms and medication changes; a classifier labelled each 7-day window prodromal or stable
METRIC	AUROC 0.68 with all three vital signs combined (SpO_2 alone 0.66, pulse rate alone 0.58)
LESSON	**Labels are choices: a different definition of an exacerbation would give the model different labels to learn from. Vital signs carry a real but modest early signal.**

Source: Shah SA et al., J Med Internet Res, 2017. DOI: 10.2196/jmir.7207

Risk stratification of COPD patients in primary care

Selecting the cohort from linked routine GP records



37,485

patients with COPD in the final cohort (from 4.78 million in the database)

23.6%

died within 10 years of follow-up

Shah SA et al., npj Primary Care Respiratory Medicine, 2022. DOI: 10.1038/s41533-022-00280-0

Case study 2: summary

Routine GP records flagged high-risk COPD patients at national scale

PROBLEM	Which COPD patients in primary care are most likely to die within 10 years, and why?
DATA	Linked CPRD primary care records: 37,485 COPD patients, 149,196 person-years of follow-up; 23.6% died within 10 years
MODEL	Cox proportional hazards model using comorbidities, symptoms, deprivation and lifestyle recorded over time
METRIC	Hazard ratios: heart failure about 2.0; ischaemic heart disease and current smoking about 1.5
LESSON	**Routinely collected data can stratify risk at scale, but coding practice shapes what the model sees.**

Source: npj Primary Care Respiratory Medicine, 2022. DOI: 10.1038/s41533-022-00280-0

Vital sign monitoring during triage in hospitals

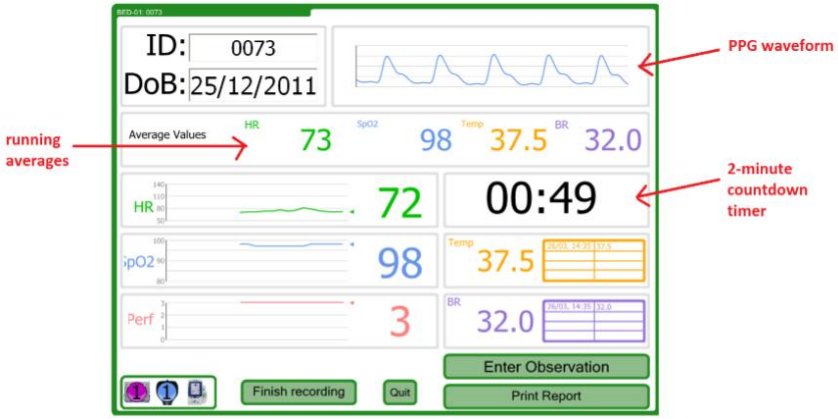
The Oxford Paediatric Emergency Department study



(a) Nonin Probe



(b) Tablet PC



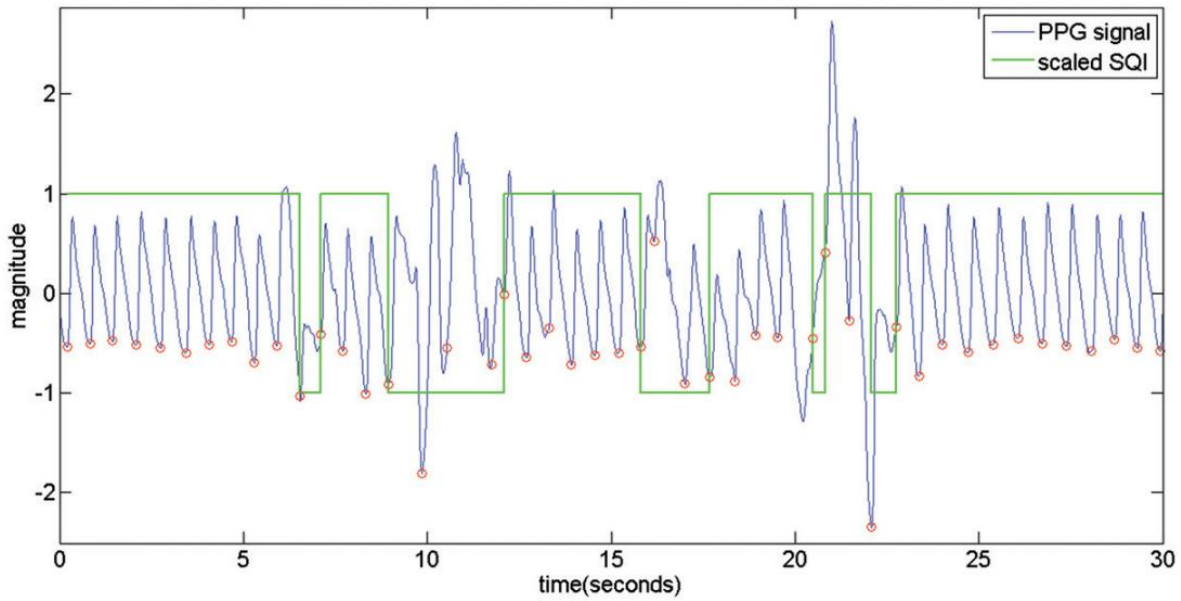
274

children presenting to the Paediatric ED were recruited

Current practice of respiratory rate estimation is based on manual counting.

Shah SA et al., Journal of Medical Engineering & Technology, 2015. DOI: 10.3109/03091902.2015.1105316

Estimating respiratory rate from the pulse oximeter signal



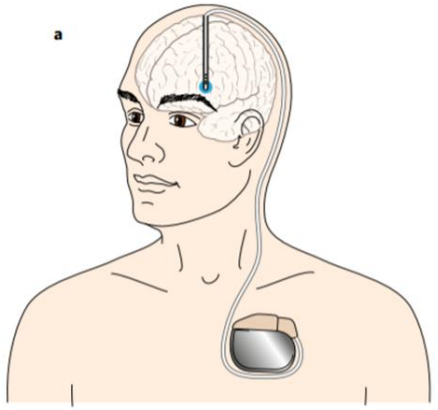
Method	Mean absolute error		Median absolute error	
	0–5 years ($n = 126$)	5–12 years ($n = 79$)	0–5 years ($n = 126$)	5–12 years ($n = 79$)
$AR_{original}$	7.9	13.2	5.9	12.5
$AR_{spectrum}$	7.5	10.8	5.7	7.2
$AR_{medianspectrum}$	7.6	5.3	6.0	4.1

Mean and median absolute error of the estimates, by age group

First study at the time to validate respiratory rate estimation from 200+ children during routine triage in a hospital.

Deep brain stimulation for dystonia in children

Can we predict which children are likely to benefit?



Deep brain stimulation involves the surgical implantation of electrodes into specific areas of the brain to modulate abnormal neural activity.

Application of Machine Learning Using Decision Trees for Prognosis of Deep Brain Stimulation of Globus Pallidus Internus for Children With Dystonia

Syed Ahmar Shah[1,2,3], Peter Brown[2,3], Hortensia Gimeno[4,5], Jean-Pierre Lin[4,5†] and Verity M. McClelland[4,6†]*

DOI: 10.3389/fneur.2020.00825

103

children with dystonia who underwent DBS

Severity scored with the BFMDRS at baseline and after 1 year

Question: which children are likely to have a favourable prognosis after DBS?

Why a decision tree? Interpretable models for small data

Disease severity | Sex | Aetiology | Imaging | SEP | CMCT

Machine learning: decision tree

Prognosis for DBS

Linear features

Inadequate if the relationship is not linear

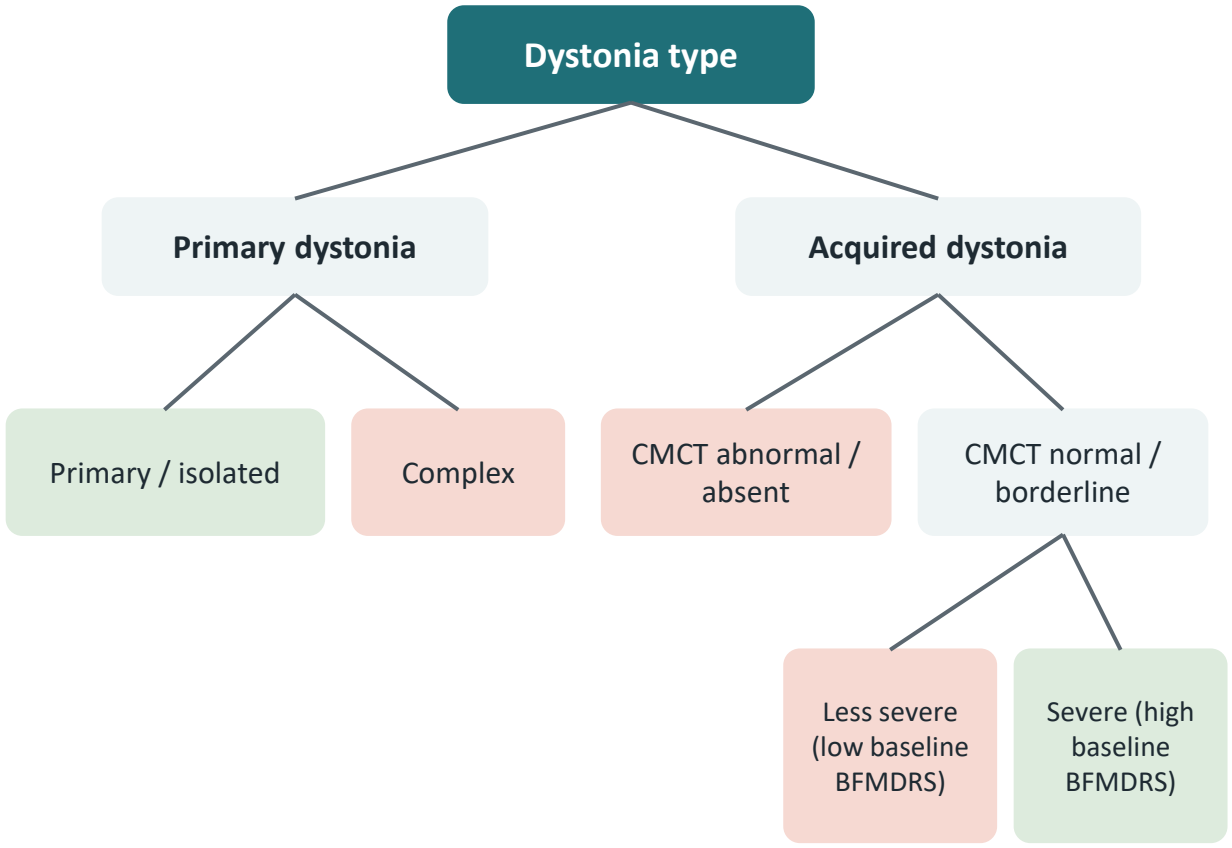
Polynomial features

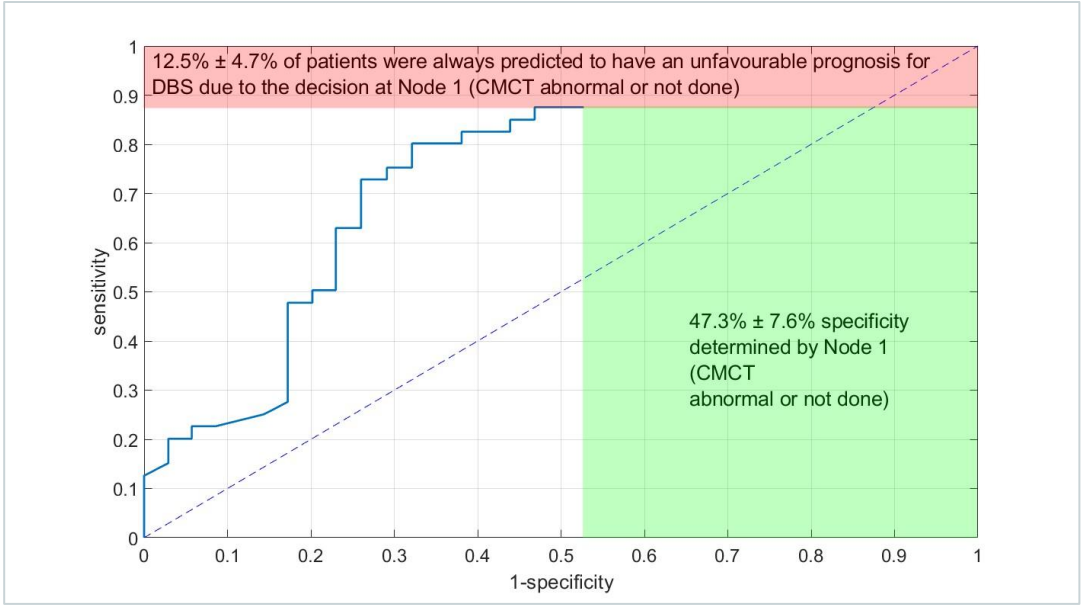
Overfits and is less interpretable

Decision tree

Interpretable, insightful, suited to categorical features

Result: a tree clinicians can read, and its ROC analysis





Favourable prognosis

Unfavourable prognosis

Case study 4: summary

A simple decision tree helps predict which children benefit from DBS

PROBLEM	Deep brain stimulation is invasive and outcomes vary. Which children with dystonia are likely to benefit?
DATA	103 children who underwent DBS, with BFMDRS severity scores at baseline and after one year
MODEL	Decision tree using dystonia type, central motor conduction time (CMCT) and baseline severity
METRIC	ROC analysis: the CMCT split at Node 1 alone gave 47% specificity (± 8%)
LESSON	**With around 100 patients, a model clinicians can read and challenge is critical and important.**

Source: Frontiers in Neurology, 2020. DOI: 10.3389/fneur.2020.00825

From understanding disruption to planning recovery

The same pandemic data supported three different kinds of decision

UNDERSTAND

What happened?

- Impact of COVID-19 on non-COVID care across the UK
- Identified the scale and persistence of disruption

eClinicalMedicine, Dec 2021. DOI: 10.1016/j.eclinm.2022.101462

SUPPORT DECISIONS

What is likely to happen next?

- Forecasting COVID-19 cases and hospitalisations
- Outputs used to inform policy decisions, such as the timing of lifting restrictions

Lancet Digital Health, Sep 2021. DOI: 10.1016/S2589-7500(21)00175-8

PLAN RECOVERY

What capacity is required?

- Elective care backlog modelling for England and Scotland
- Scenario analysis of the capacity increases needed
- Policymaker engagement and public dissemination

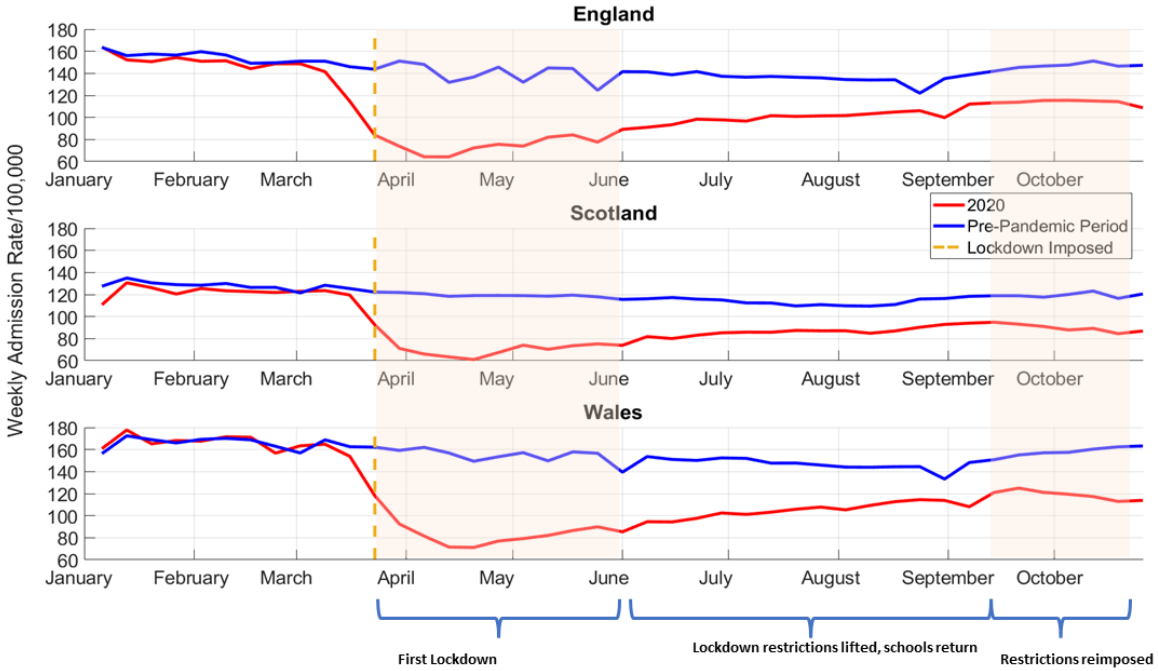
Lancet, Jan 2024. DOI: 10.1016/S0140-6736(23)02744-7

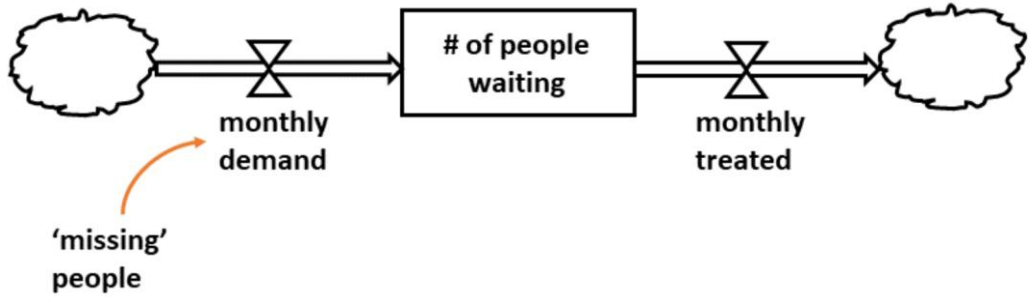
Lancet Reg Health Eur, 2025. DOI:10.1016/j.lanepe.2024.101188

Modelling can directly inform system-level decisions and policy.

Disruption and recovery of hospital services

Modelling the waiting list as a stock with monthly flows





A forecasting model of the waiting list

- The list depends on its own recent past
- It also depends on demand (new referrals) and on treatment capacity
- So we can ask: what happens if capacity changes?

Hospital admissions fell sharply after the first lockdown (England, Scotland, Wales)

Case study 5: summary

NHS Scotland needs 20% more elective capacity over three years

PROBLEM	How large is the COVID-19 backlog in elective care, and how much extra capacity would clear it?
DATA	Public Health England and Scotland monthly waiting-list statistics, 2013–2023, at national, regional and specialty level
MODEL	Vector autoregression with exogenous variables (VARX), projecting the waiting list under capacity scenarios
METRIC	667,000 referrals waiting at end of 2023; with no extra capacity, nearly 1 million by end of 2026 (Scotland results)
LESSON	**System models inform policy rather than individual care. Their value is in comparing scenarios, not a single forecast.**

Sources: Shah SA et al., Lancet Reg Health Eur, 2025. DOI: 10.1016/j.lanepe.2024.101188; Lancet, 2024. DOI: 10.1016/S0140-6736(23)02744-7

What the headlines picked up

The Scotsman front page, 10 January 2025



~1 million

people on the waiting list by the end of 2026 if capacity does not increase: the figure in the headline

What we modelled

Several forecasts and capacity scenarios, including how much extra capacity would clear the backlog.

What the headline kept

One number, in a way that was easy to remember and repeat.

The Scotsman, 10 January 2025. Study: Shah SA et al., Lancet Reg Health Eur, 2025. DOI: 10.1016/j.lanepe.2024.101188

You never know what gets picked up

Several forecasts and scenarios went in; a few headlines came out



- **Several forecasts and scenarios**
- Outlets chose different numbers and different framings
- **Once published, you do not control the story**
- Give the assumptions with the numbers, and say what they do not show

Expect the simplest number to be the one that travels. Make sure it can carry its caveats.

Press coverage: The Scotsman and The Herald, 10 January 2025.

Asthma exacerbations during the COVID-19 pandemic

A cohort study with an interrupted time-series design

Cohort study of patients with asthma in a national English primary care database

Patient identification period	Follow-up: before lockdown	Lockdown
1 Jan 2010 to 31 Dec 2015	31 Dec 2015 to 24 Mar 2020	24 Mar to 15 Aug 2020

Design

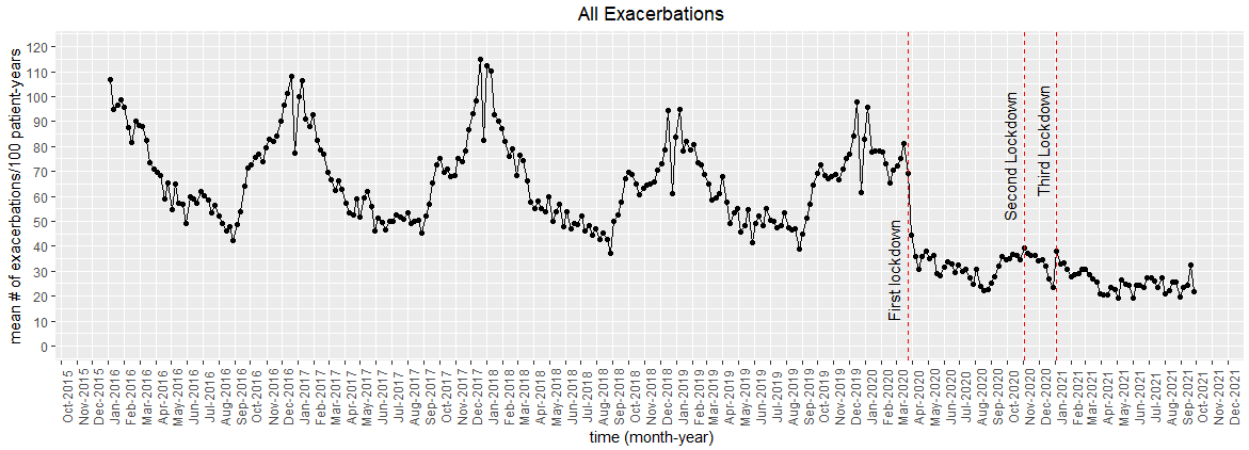
Interrupted time series: compare the trend after the intervention date with the trend expected from before.

Why it matters

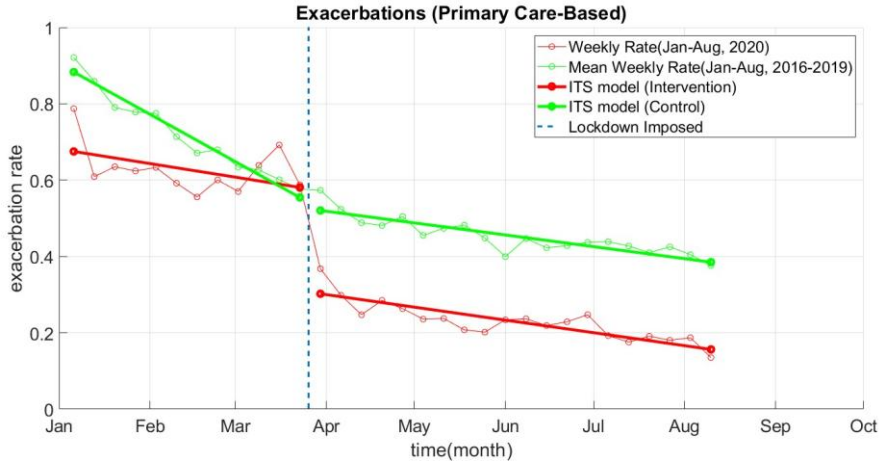
Natural experiments let us ask what changed when behaviour and care changed overnight.

Shah SA et al., Thorax, 2021. DOI: 10.1136/thoraxjnl-2020-216512; Lancet Reg Health Eur, 2022. DOI: 10.1016/j.lanepe.2022.100428

Results: exacerbations in primary care fell after lockdown



Asthma exacerbations in primary care over time

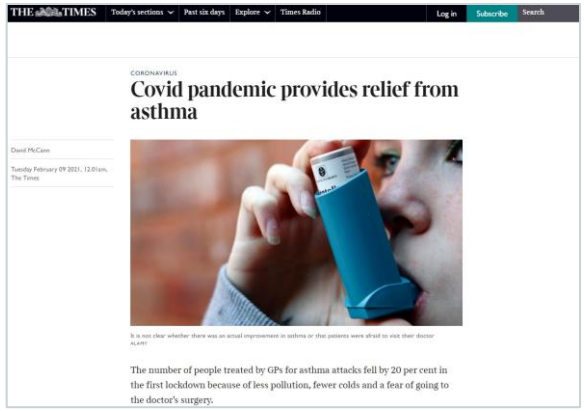


Interrupted time-series analysis: observed vs expected

A drop in recorded exacerbations is a finding. Why it happened is a separate question.

One study, opposite explanations: why?

The same results, reported in the press





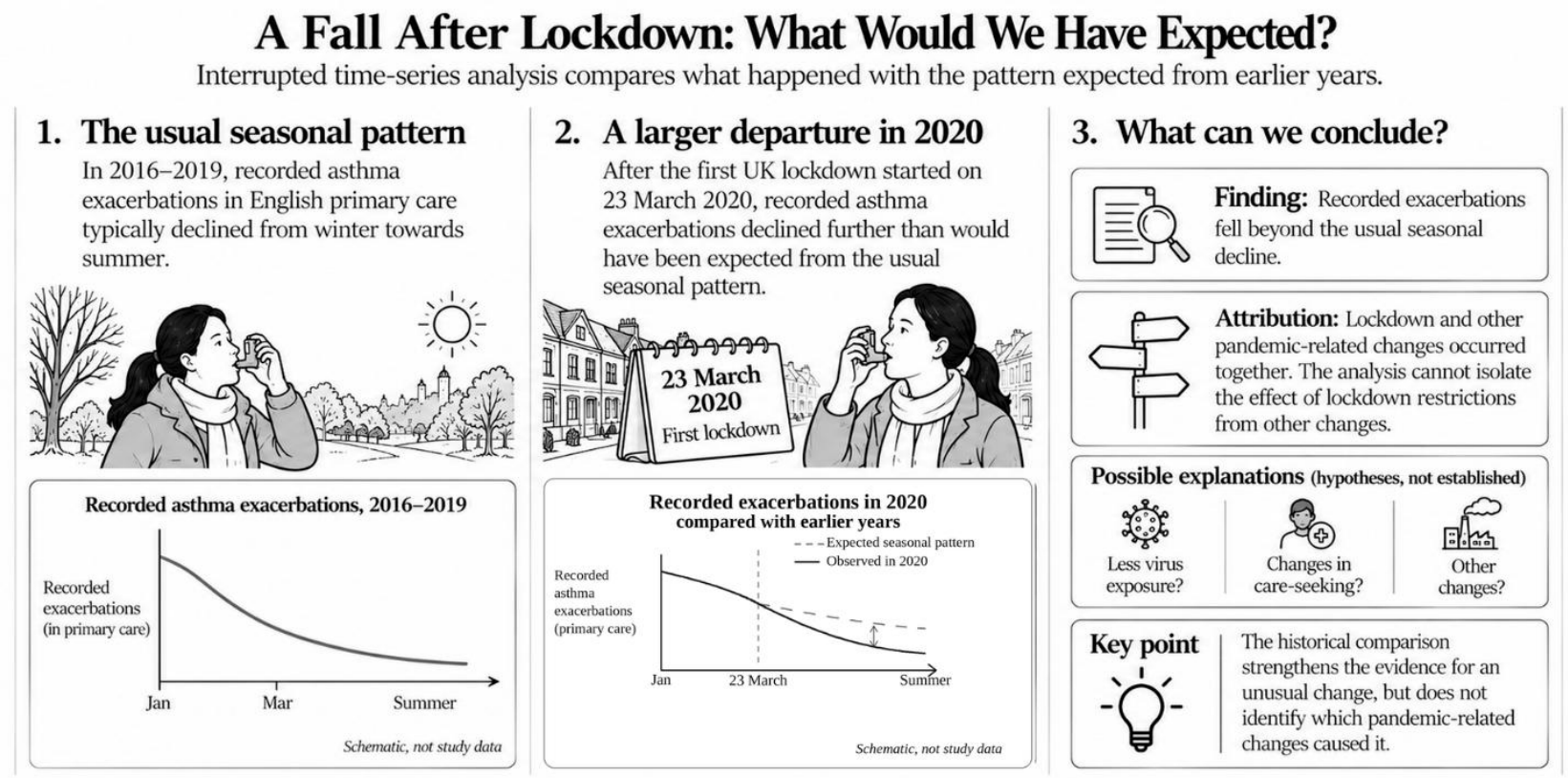




Same study, diametrically opposite explanations. What would you need to know to decide between them?

A fall after lockdown: what would we have expected?

Comparing what happened with what would have happened



A prediction or a trend comparison supports a finding. A causal claim needs more evidence.

Figure from: Shah SA. Understanding AI for the Real World (in preparation). Data: Shah SA et al., Lancet Reg Health Eur, 2022.

What we covered

Seven studies, the data they used and the models behind them

CASE STUDY	DATA	MODEL TYPE	MAIN LEVEL
1 COPD home monitoring	Digital monitoring: SpO_2, pulse and respiratory rate, symptom diary	Classifier (supervised machine learning)	Patient
2 COPD risk in primary care	Linked routine GP records (CPRD)	Cox proportional hazards model	Patient and population
3 Respiratory rate in triage	Pulse oximeter signals from a children's emergency department	Signal processing and estimation	Patient
4 Deep brain stimulation	Clinical and neurophysiology data from 103 children	Decision tree	Patient
5A Pandemic disruption and forecasts	National and UK-wide health service and case data	Interrupted time series; forecasting	Population and system
5B Waiting-list backlog	Public Health England/Scotland monthly statistics	Vector autoregression scenarios (VARX)	System
6 Asthma and lockdown	Routine primary care records, national English database	Cohort study; interrupted time series	Population

The levels overlap, and the purpose is the same: to inform care for everyone. The question comes first.

4

Can we trust it? Data, evidence and ethics

“Data does not equal information; information does not equal knowledge; and, most importantly of all, knowledge does not equal wisdom.”

A number on the screen: the pulse oximeter

An instrument can show a clear, precise-looking number while concealing an error



It estimates; it does not count. Red and infrared light pass through the finger. Oxygenated and deoxygenated haemoglobin absorb them differently, and the device converts that into an oxygen saturation estimate.

It is genuinely useful. Fast, non-invasive and repeatable, and a clinician can read it alongside symptoms and other observations.

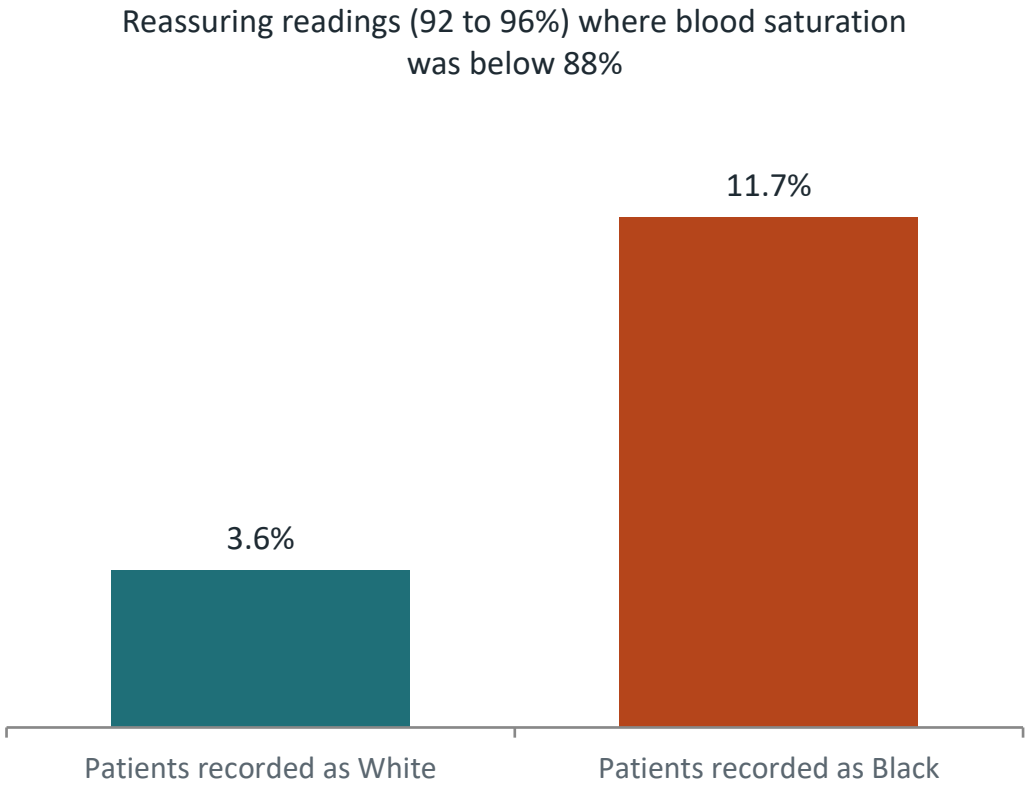
The question is not whether it is useful. It is when, for whom and for what purpose the estimate is dependable enough.

The same device generated the data in two of my case studies.

Shah SA. Understanding AI for the Real World (in preparation), chapter: Data Are Not Reality

The same reassuring reading, a different truth

When the oximeter said 92% to 96%, how often was true saturation below 88%: in Black vs White patients?



Black vs White patients: about three times as often (11.7% vs 3.6%)

A comfortable number on the screen, while the blood told a more worrying story.

The science and regulators responded

A COVID-19 study of 7,000+ patients confirmed it; the UK equity review (2024) and FDA draft guidance (2025) followed.

The study had limits

Selected hospital measurements and recorded race categories, not measured skin pigmentation: it showed what, not why.

An error that is neither obvious nor evenly distributed can hide inside a precise-looking number.

Sjoding MW et al., NEJM 2020;383:2477-2478. Fawzy A et al., JAMA Intern Med 2022. UK DHSC Equity in Medical Devices review, March 2024. FDA draft guidance, January 2025.

Different data problems require different remedies

Recorded data can depart from reality for different reasons, so each problem requires a different response.

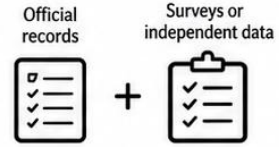
1 Unrecorded events

Some events happen but are never observed, reported or recorded.



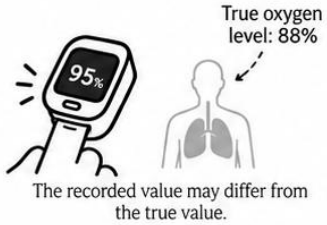
Use complementary data sources

Combine multiple sources, such as surveys, audits or other datasets, to reveal what individual records miss.



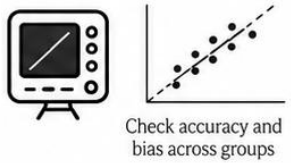
2 Measurement error

Recorded values can be inaccurate, even when measurements are consistent.



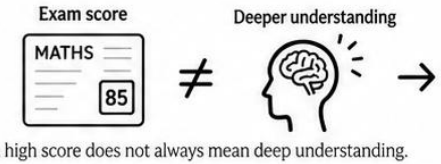
Validate measurements

Compare with independent reference standards, improve instruments and assess performance across different groups and settings.



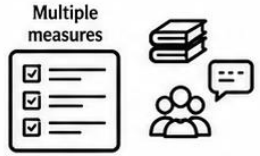
3 Proxy or label mismatch

Recorded labels or measures may not fully capture the concept of interest.



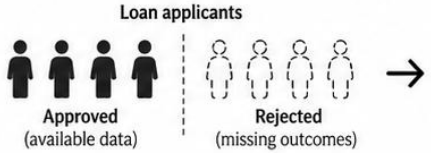
Examine validity of measures and labels

Consider whether recorded variables adequately represent the intended construct, and use better measures or additional evidence where possible.



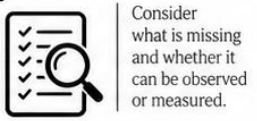
4 Selection bias

The data may include only a selected group, due to previous decisions or processes.



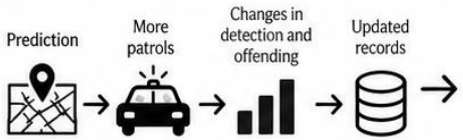
Examine selection and seek additional evidence

Understand how people or events enter the data, state assumptions and, where feasible, use additional data or study designs that generate new evidence.



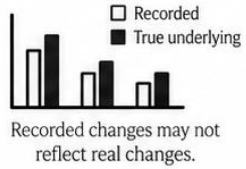
5 Feedback-shaped data

Predictions and decisions can change future observations, affecting what is recorded.



Assess feedback using independent evidence

Consider how predictions and decisions affect subsequent data, and use independent evaluation, such as control areas or pre-specified studies, to separate detection from real changes in underlying events.



Better evidence requires understanding the problem, not simply collecting more data.

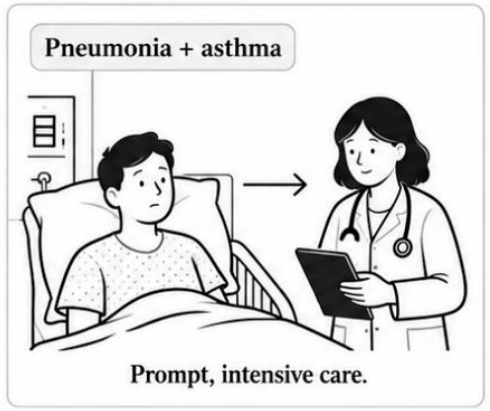
Figure from: Shah SA. Understanding AI for the Real World (in preparation). Example: Cooper et al. 1997; Caruana et al. 2015

A Predictor Is Not Necessarily a Lever

The pneumonia example: a pattern in the data can be shaped by the treatment patients received.

1. Care in the real world

A treatment policy shapes outcomes.

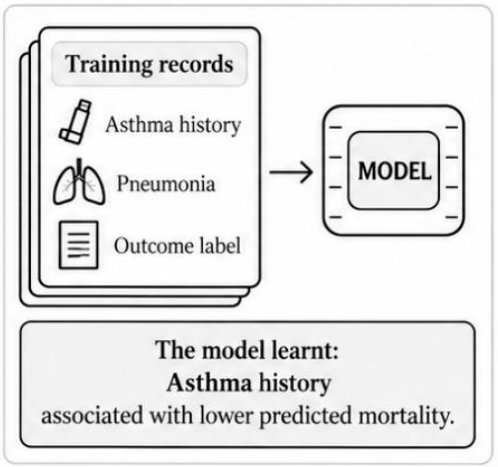


In the hospitals studied, patients with asthma often received more aggressive care for pneumonia.

Care shaped the recorded outcomes.

2. Pattern in the data

A model learns the observed association.

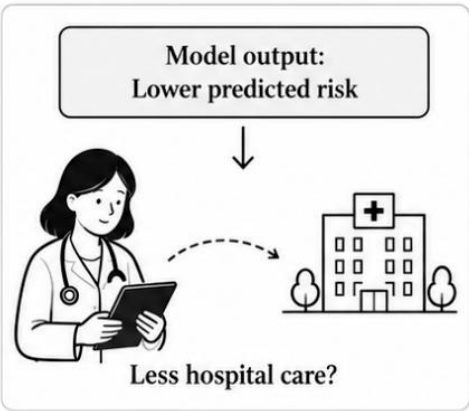


The pattern was real in these records, under the care patients received.

Association is not protection.

3. If the model guides care

Changing practice can change the pattern.



Using the association to reduce care could remove the very help that shaped the observed outcome.

The prediction may no longer hold.

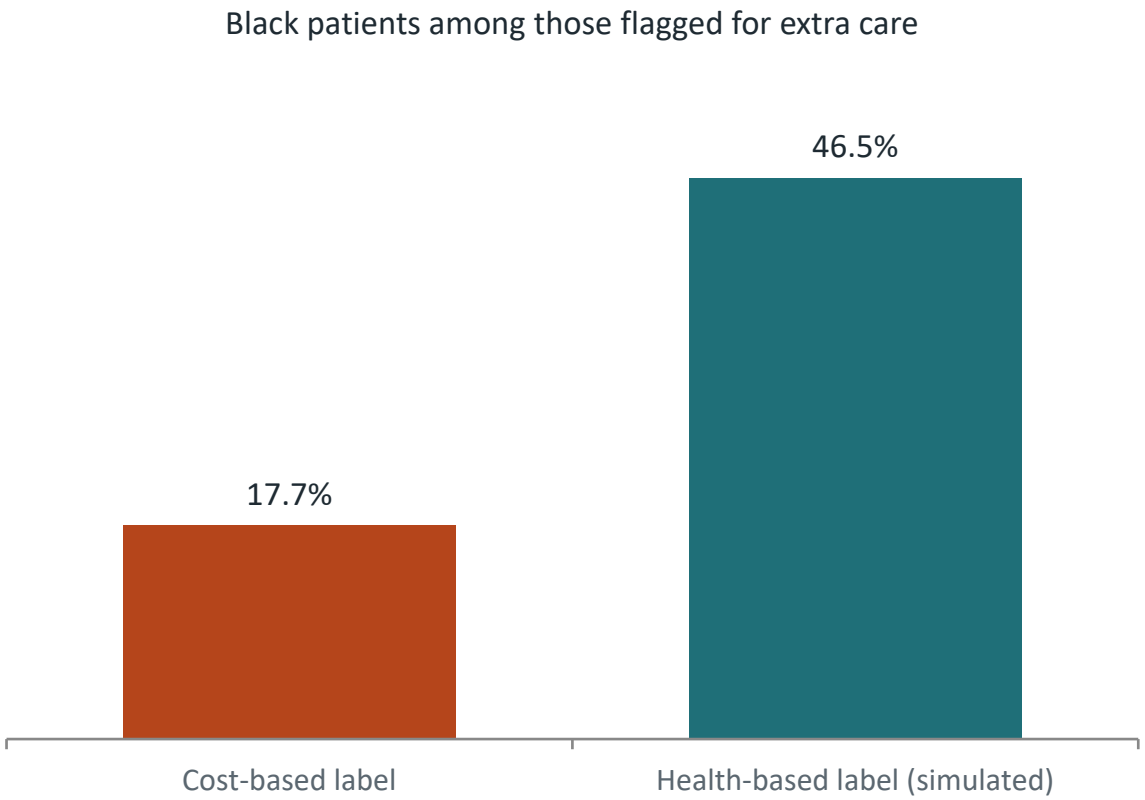
Observed outcomes reflect existing care.

A model may predict those outcomes well without showing what happens if that care changes.

Figure from: Shah SA. Understanding AI for the Real World (in preparation). Example: Cooper et al. 1997; Caruana et al. 2015

Bias through the label

A cost-based label under-referred Black patients by more than half



GOAL

Find patients who need extra care

LABEL USED

Predicted future healthcare cost

HIDDEN PROBLEM

Less is spent on Black patients with the same health needs

RESULT

At the same risk score, Black patients were sicker

Check what your label actually measures. In Case Study 1, the model learned from our definition of an exacerbation; a different one would have changed what it learned.

Obermeyer Z et al., Dissecting racial bias in an algorithm used to manage the health of populations. Science, 2019 (health-based label: simulated)

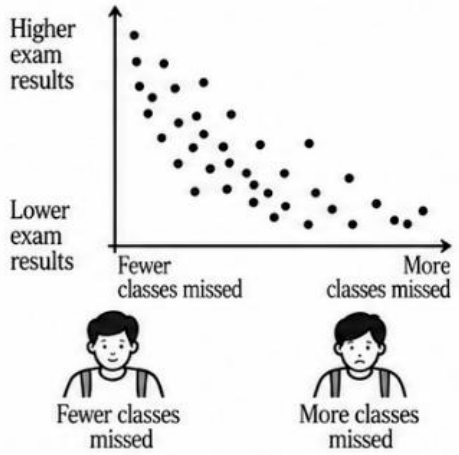
Four Claims Need Different Evidence

Association, prediction, causal effect and mechanisms answer different questions. Each requires different evidence.

1. Association

Are X and Y related in the observed data?

Example:
Do pupils who miss more classes tend to have lower examination results?

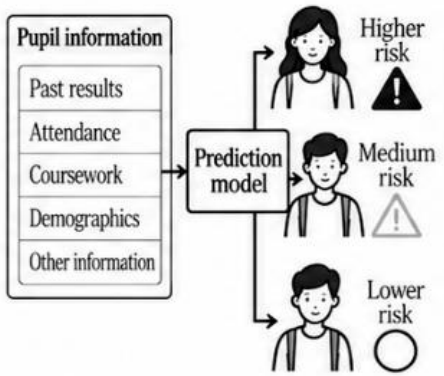


Evidence needed
Observed patterns in recorded data.

2. Prediction

Who is likely to have lower examination results?

Example:
Using past attendance and other information to predict who may struggle.

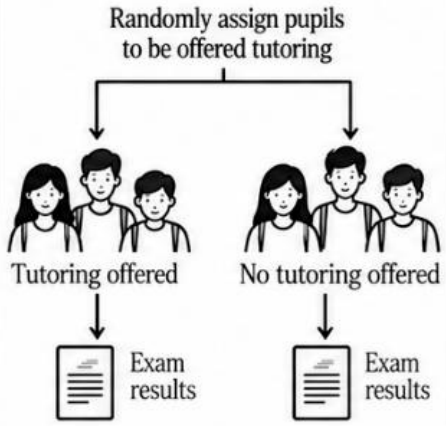


Evidence needed
Test the predictions on new data.

3. Causal effect

Would offering tutoring change examination results?

Example:
Does offering tutoring improve examination results?



Evidence needed
Randomised experiments, or carefully designed observational comparisons.

4. Mechanism

How might offering tutoring improve examination results?

Example:
If offering tutoring improves results, what might explain it?



Evidence needed
Studies of possible pathways and intermediate changes, alongside outcome evidence.

Prediction is not explanation. Different claims require different kinds of evidence.

Figure from: Shah SA. Understanding AI for the Real World (in preparation). Example: Cooper et al. 1997; Caruana et al. 2015

External validation: does it work somewhere new?

A widely used sepsis model fell to AUROC 0.63 at a new hospital

0.76–0.83

AUROC reported by the developer

0.63

AUROC at Michigan Medicine, 38,455 hospitalisations

At the deployment threshold, the model:

67%	of sepsis cases missed
18%	of all inpatients triggered an alert
12%	of alerts were true sepsis (PPV)

Test on a different hospital, population or time period before deployment. Developer-site performance is usually the best case.

Wong A et al., External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. JAMA Internal Medicine, 2021;181(8):1065-1070

Six stages from model to bedside

Most published models never get past the first stage

1	**Develop and test internally**	Performance on held-out data from the same source	**TRIPOD+AI**
2	**Validate externally**	Performance at a new site, population or period	**TRIPOD+AI**
3	**Evaluate in early clinical use**	Silent or small-scale live use: workflow, safety, trust	**DECIDE-AI**
4	**Test the impact**	Does using it improve decisions and outcomes?	**CONSORT-AI, SPIRIT-AI**
5	**Gain regulatory approval**	Medical device rules and AI regulation	**MDR, EU AI Act**
6	**Monitor and update**	Track drift, recalibrate, re-validate after changes	**Post-market surveillance**

Most published models never get past stage 1. Evidence has to climb the ladder before impact is claimed.

The AI chasm: strong models are not real-world impact

We are very good at building models, much less good at changing decisions and outcomes

731

COVID-19 prediction models identified in a BMJ review (2020)

about 90% not suitable for further investigation

86

prospective studies among about 10,000 identified in a Lancet Digital Health review (2024)

of AI studies that influence decision-making

Where does it break?

Models are built

but not implemented

or not used

or fail to improve outcomes

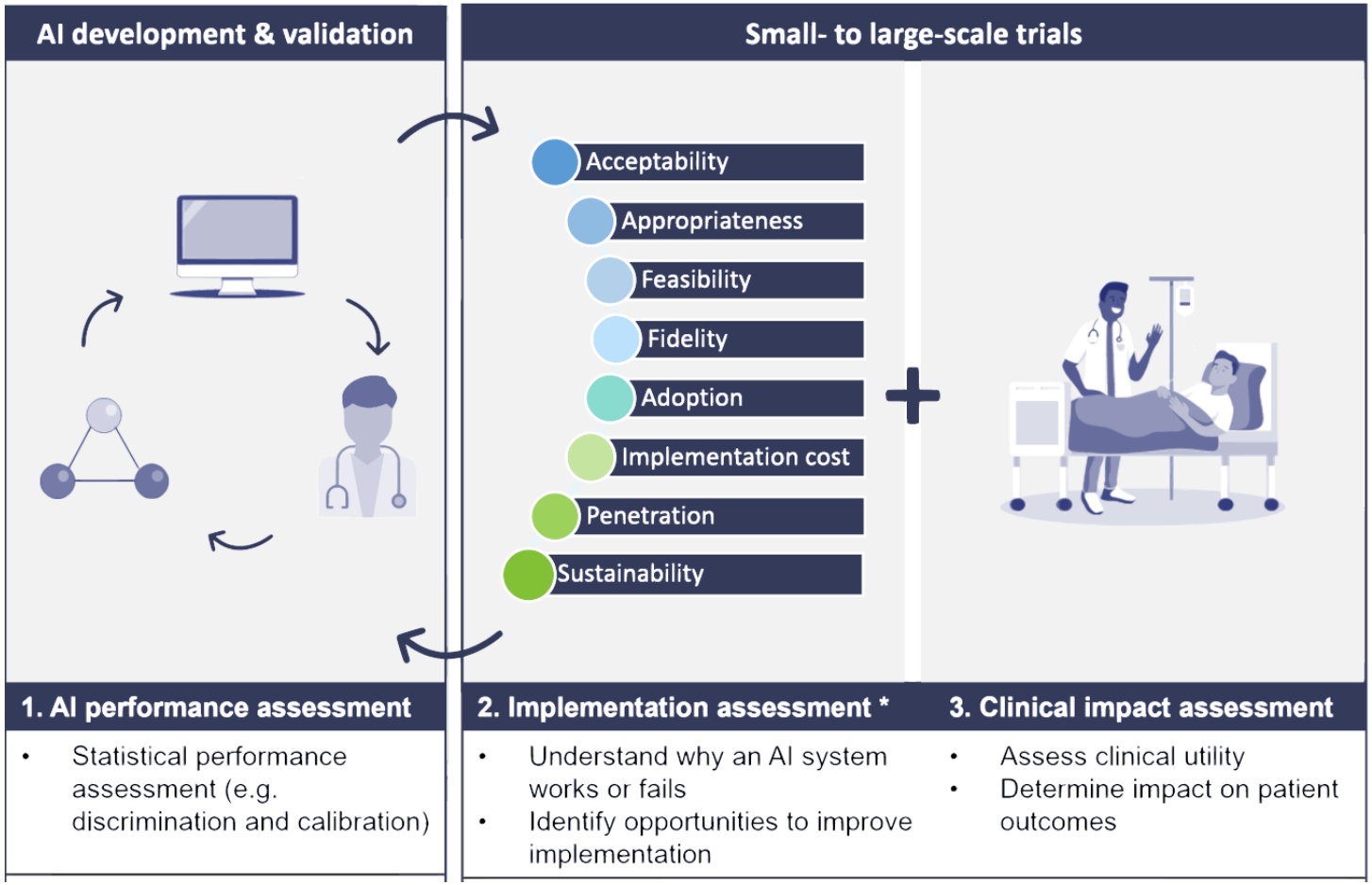
Approval does not guarantee use, and use does not guarantee impact.

Failure can occur at any step: data, technical, workflow, behavioural or system constraints.

Wynants L et al., BMJ 2020 (DOI 10.1136/bmj.m1328); Lancet Digital Health 2024 (DOI 10.1016/S2589-7500(24)00047-5)

The way we assess models will need to change

Beyond performance metrics: implementation outcomes



Eight implementation outcomes

Acceptability	Appropriateness
Feasibility	Fidelity
Adoption	Implementation cost
Penetration	Sustainability

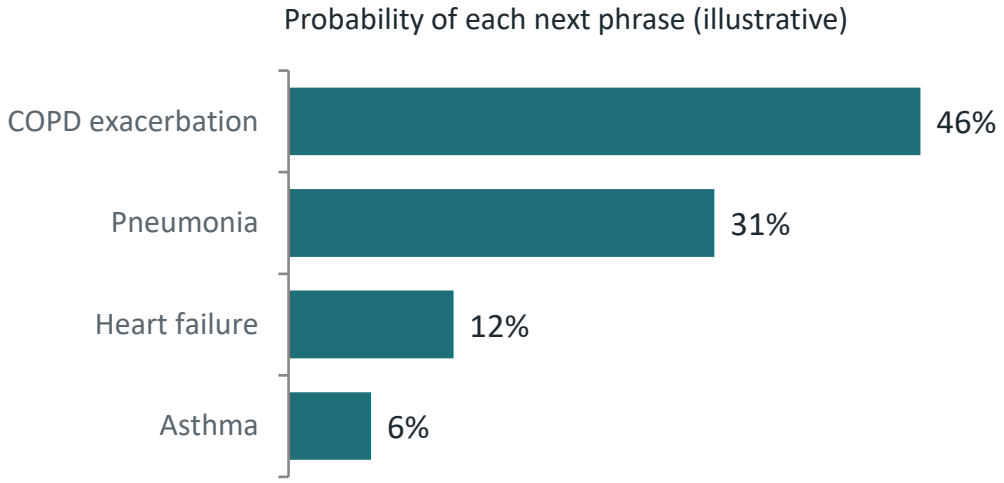
Framework from: Nature Medicine, January 2022. DOI: 10.1038/s41591-021-01614-0

How large language models work

They predict the next word, so fluency is not evidence of correctness

PROMPT

"68-year-old smoker, three days of worsening breathlessness and green sputum. The most likely cause is …"



Trained to predict

Learns from vast amounts of text to predict the next word (token).

Tuned to help

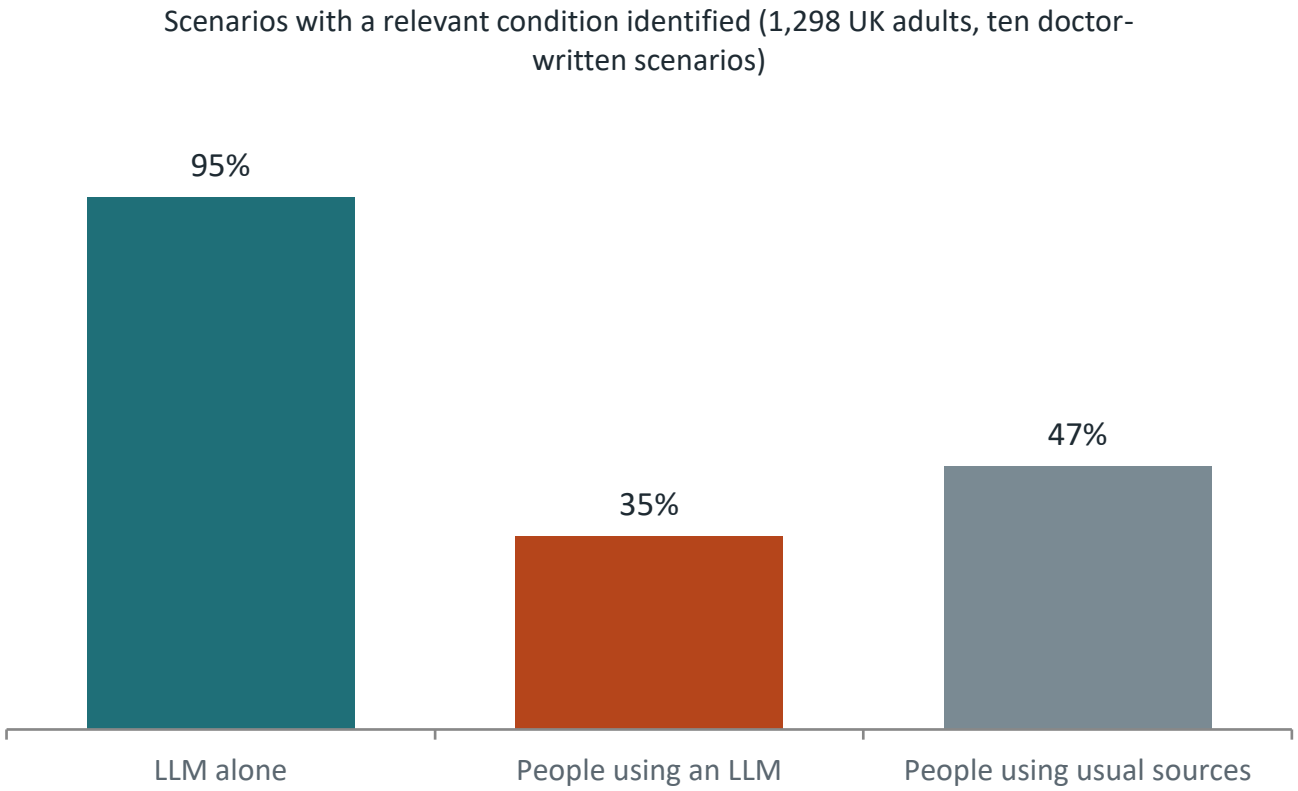
Further trained to follow instructions and sound helpful.

Not checked against truth

Nothing in the process verifies facts, so errors can sound confident.

Benchmarks vs real use

LLMs alone found the condition 95% of the time; people using them, 35% at best



Why the gap?

- People left out key details when describing symptoms
- The right condition appeared in about 70% of chats, but users often did not act on it
- Small wording changes shifted the model's answer

Test with real users before deployment. Benchmark scores did not predict how people fared.

"LLM alone" averages GPT-4o, Llama 3 and Command R+; "people using an LLM" is the best of the three groups. Source: Bean AM et al., Nature Medicine, 2026;32:609-615

Clinical LLM research: publications vs clinical evidence

Peer-reviewed studies published January 2022 to September 2025

4,609
peer-reviewed studies identified

1,048
estimated studies using real-world patient data

19
identified prospective randomised trials

100%
of the studies identified

about 23%
of the studies identified

under 0.5%
of the studies identified

A growing publication literature is not the same as a mature evidence base for clinical practice.

Chen et al., Nature Medicine (2026). DOI: 10.1038/s41591-026-04229-5

Six ethical questions to ask of any health AI system

A checklist you can apply to any health AI system, including the studies in this lecture

Fairness

Does it work equally well for every patient group?

Validity

Has it been tested somewhere new, and does using it improve outcomes?

Transparency

Can clinicians and patients understand and challenge it?

Privacy and consent

Whose data trained it, and what did they agree to?

Accountability

Who is responsible when it is wrong?

Safety over time

Who monitors it as populations and practice change?

Good enough to support which claim, about which population, for which decision, and compared with what alternative?

5

The future: opportunities and challenges

What makes healthcare unique, and what do we take away?

What makes healthcare innovation unique?

After 15+ years in healthcare, these are my top four reasons innovation is slow to be adopted

Strong regulatory barriers

Safety must be shown before use, and evidence takes time.

Data access challenges

Data are fragmented across settings, formats and organisations.

Real-world complexity

People, workflows and context decide whether a tool is used.

Interconnected ecosystem

A change in one part of the system changes others.

The features that make healthcare worth transforming also make it hard to change.

Will AI replace doctors?

A personal reflection



"I can very easily foresee an AI algorithm 'telling' you that a child has a serious infection that combines various pieces of information. But then what?"

You still need a decision-maker at the other end who needs to consume that information and make a decision.

As a parent, I can only take comfort in the fact that the decision-maker is a human being, trained and with the responsibility, compassion and intelligence to see beyond the confines of a closed system trained with pre-determined inputs.

Discuss: what decisions should always stay with a human, and why?

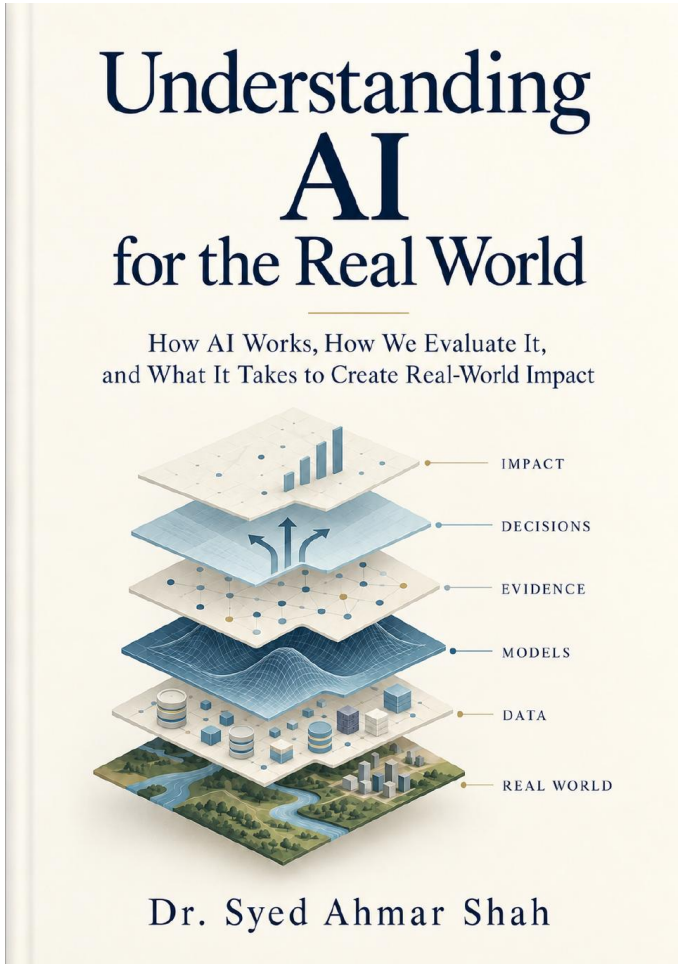
Adapted from a personal LinkedIn post by the speaker

Five things to take away

1	**The question comes first**	Choose data and methods to fit the question, not the other way round.
2	**Data are not reality**	Every dataset is a partial record shaped by how things were measured, selected and labelled.
3	**Metrics need context**	AUROC, sensitivity and PPV only mean something with prevalence, calibration and a decision in mind.
4	**Test somewhere new**	Validate externally, evaluate in use, and keep monitoring after deployment.
5	**Ethics is part of the method**	Ask who it works for, who is accountable, and whether it improves outcomes.

If you want to learn more

My book in progress follows AI from the real world to its impact



Understanding AI for the Real World. How AI works, how we evaluate it, and what it takes to create real-world impact.

Part I: Foundations, how AI works.

Part II: Evidence and reliability, from data to reliable AI.

Part III: Decisions and impact, from AI outputs to real-world impact.



Questions?

Dr Syed Ahmar Shah